

Final Year Project
on
**Flow Rate Measurement of Self-Compacting
Concrete Using Chirped Fiber Bragg
Grating Sensors**

Submitted in partial fulfillment of the requirements of the degree of
Bachelor of Technology by

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Abstract

This paper presents the design and analysis of a chirped fiber Bragg grating (CFBG) sensor to monitor concrete flow dynamics in real-time basis for the first time. The proposed sensor utilizes wavelength-encoded strain variations to measure pressure changes, enabling non-intrusive and highly sensitive detection. Traditional sensors often face challenges such as electromagnetic interference and signal degradation, which the proposed CFBG sensor overcomes through its broad reflection spectrum and enhanced detection mechanism. The performance of the proposed CFBG is analyzed by considering the parameters such as sensitivity and figure of merit (FOM). In order to achieve high performance of the proposed sensor in a noisy environment, a High Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP) method is used in this paper. The proposed system achieves a strain sensitivity of $2 \text{ pm}/\mu\epsilon$ with a FOM of $0.01 \mu\epsilon^{-1}$, ensuring high precision. Moreover, the flowrate sensitivity of the proposed sensor is calculated using strain sensing data and obtained as $0.006454 \text{ nm}/(\text{m}^3/\text{s})$. In addition, the proposed sensor exploited a temperature compensation mechanism using dual FBG sensor system to measure the strain accurately. It has been found from the simulation results that the proposed sensor can be well-suited for real-time concrete flow monitoring, which provides a promising solution for structural health monitoring in civil construction applications.

Index Terms— Chirped Fiber Bragg Grating (CFBG), Concrete Flow Monitoring, Strain Sensor, HMNS-PP, FOM, Real-Time Sensing, Temperature Compensation.

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Chapter 1

1.1 Introduction

Structural Health Monitoring (SHM) has emerged as an essential domain in modern civil engineering, offering a comprehensive framework for assessing and ensuring the integrity, safety, and longevity of critical infrastructures. Among various components used in construction, concrete stands out as a fundamental material whose quality during placement plays a pivotal role in determining the structural soundness of the entire framework. The dynamics of concrete flow during pouring can significantly affect the internal uniformity of the structure. Irregularities such as incomplete filling, segregation of aggregates, and air voids often arise from uncontrolled or unmonitored concrete flow, leading to structural vulnerabilities and compromised performance over time.

The development and widespread use of self-compacting concrete (SCC) has further elevated the need for precise monitoring systems. SCC is designed to flow under its own weight, filling intricate mold geometries and densely reinforced areas without the need for mechanical vibration. However, this fluid-like behavior also introduces variability in flow behavior, which if left unchecked, can lead to uneven compaction or differential pressure zones within the formwork. Thus, monitoring pressure dynamics during the concrete placement process becomes critical to ensure the material maintains its intended properties and structural homogeneity.

Traditional pressure sensing techniques such as piezoelectric and electrical sensors have been deployed for civil infrastructure monitoring. While effective to an extent, these technologies suffer from significant drawbacks, particularly in harsh or electromagnetically noisy environments. Their susceptibility to signal degradation, electromagnetic interference (EMI), and reduced accuracy in long-term deployments limits their reliability in real-time construction applications. Additionally, scalability becomes a major issue when large areas or complex geometries need to be instrumented.

Fiber Bragg Grating (FBG) sensors, in contrast, have gained recognition due to their superior performance in harsh environments. These optical sensors operate by reflecting specific wavelengths of light that shift in response to strain or temperature changes. Their immunity to EMI, ability to function in chemically and thermally unstable settings, and ease of integration into distributed networks make them ideal for applications in SHM. Previous works, such as those by Sahota et al., have elaborated on the underlying principles of FBGs, particularly the Bragg wavelength dependence on strain and refractive index variations, forming the foundation for pressure and strain sensing.

Further extending the capabilities of standard FBGs, chirped fiber Bragg gratings (CFBGs) offer advanced features, particularly suited for pressure sensing applications. Unlike uniform FBGs, which maintain a constant grating period and are generally suited for static strain measurements, CFBGs have a gradually varying grating period along their length. This results in a broader and more complex reflection spectrum, which can be used to monitor distributed strain variations over a spatial region. Studies by Tosi and others have reviewed the applications and mathematical modeling of CFBGs, establishing their significance in dynamic sensing environments. Compared to other configurations such as tilted FBGs—which offer multiparameter sensing but suffer from high insertion loss and mechanical fragility—CFBGs present a robust and high-sensitivity alternative.

The application of CFBG in fluidic environments, including concrete flow scenarios, has been validated through experimental models such as those performed by Koyama et al., where relationships between pressure, viscosity, and induced strain were explored using a simulated blood flow system. The insights from such studies support the feasibility of adapting CFBGs for monitoring pressure gradients during concrete pouring. When embedded in concrete formwork, CFBG arrays can detect strain variations induced by flowing concrete, enabling real-time visualization of flow uniformity and early detection of flow-induced defects.

One of the defining features of FBG-based systems, particularly CFBGs, is their compatibility with wavelength-division multiplexing (WDM). This allows multiple sensors to operate on a single optical fiber line without interference, which supports large-scale deployment for comprehensive structural monitoring. Despite these advantages, real-world deployment of FBG systems faces challenges due to external noise sources such as temperature fluctuations and high-frequency mechanical vibrations. These factors can obscure strain-induced wavelength shifts by introducing additional spectral changes, leading to inaccurate readings.

To mitigate the effects of thermal variations, temperature compensation strategies

have been developed. The dual-FBG method, wherein one grating senses both strain and temperature while a second isolated grating senses only temperature, is commonly used to extract pure strain information by subtracting the thermal contribution. More recent advances, such as triple-FBG configurations, further improve compensation accuracy by reducing thermal drift significantly. Hybrid sensing approaches that combine CFBGs with Fabry–Pérot interferometers (FPI) have also been proposed to achieve sub-picometer resolution in simultaneous strain and temperature measurements, demonstrating promising outcomes for dynamic environments.

However, even with these improvements, practical challenges such as spectral noise, cross-sensitivity, and distortion under high-dynamic conditions continue to limit sensing fidelity. The complex behavior of fresh concrete, characterized by heterogeneous flow and fluctuating pressure profiles, introduces additional complications. Hence, the development of advanced sensing frameworks that are not only sensitive and scalable but also noise-resistant remains a critical objective.

This project proposes a refined CFBG-based pressure sensor system tailored for real-time monitoring of concrete flow. Leveraging the inherent advantages of CFBGs—such as broad spectral bandwidth, superior sensitivity, and multiplexing capability—the system offers improved accuracy and operational reliability in construction environments. The sensor exhibits a strain sensitivity of $2 \text{ pm}/\mu\epsilon$ and flow rate sensitivity of $0.006454 \text{ nm}/(\text{m}^3/\text{s})$, validated under controlled, simulated construction conditions. These values are based on and further expand upon the experimental validation framework proposed by Yu et al., where spectral responses of CFBGs were correlated to fluidic pressure variations.

To enhance signal quality and ensure accurate peak detection in the presence of noise, a novel signal processing technique called Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP) is incorporated. Adapted from the work of Tang et al., this algorithm synergizes empirical mode decomposition (EMD) with adaptive thresholding to effectively suppress unwanted spectral noise while preserving key spectral features critical for accurate strain and pressure readings. It has been demonstrated to improve the signal-to-noise ratio by up to 18 dB, significantly boosting system robustness.

1.2 Problem Statement

Self-Compacting Concrete (SCC) enhances construction efficiency and structural durability but lacks reliable real-time monitoring of its flow properties during placement. Conventional sensors (e.g., electrical or piezoelectric) are prone to electromagnetic interference (EMI), signal degradation, and limited resolution in harsh environments, leading to undetected flow irregularities such as voids, segregation, and improper compaction. These defects compromise structural integrity and longevity. Existing fiber-optic solutions, while EMI-resistant, struggle with temperature cross-sensitivity, noise in dynamic conditions, and insufficient spatial resolution for distributed flow mapping. There is a critical need for a robust, high-fidelity sensing system that can accurately monitor SCC flow dynamics, resolve pressure gradients in real time, and mitigate environmental noise to ensure optimal concrete placement and structural performance.

1.3 Motivation

The motivation for this project stems from the growing adoption of Self-Compacting Concrete (SCC) in modern construction, valued for its ability to flow and fill formworks without the need for external vibration. While SCC offers clear advantages in terms of construction efficiency, reduced labor, and improved finish quality, its reliance on flow properties makes real-time monitoring during placement critical. Traditional assessment methods, such as slump flow and L-box tests, are manual, discrete, and prone to human error, lacking the spatial and temporal resolution needed to detect issues like segregation, incomplete filling, or improper compaction.

To overcome these limitations, this project proposes the use of a Chirped Fiber Bragg Grating (CFBG) sensor system capable of continuously tracking SCC flow behavior through strain-induced Bragg wavelength shifts. CFBG sensors are well-suited for such applications due to their high sensitivity, EMI immunity, and ability to encode spatial information. The use of chirped gratings allows for distributed sensing over longer fiber lengths, making it possible to monitor flow dynamics with fine spatial resolution.

To ensure measurement reliability in dynamic and noisy construction environments, the system incorporates a Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP) algorithm and dual-FBG-based temperature compensation. This enables the sensor to accurately distinguish strain-related spectral changes from those induced by environmental temperature fluctuations or spectral noise.

This work aims to modernize the monitoring and quality control of SCC by pro-

viding a robust, automated solution that replaces subjective manual evaluations. The developed system not only enhances the reliability and accuracy of flow rate estimation but also contributes to safer, smarter, and more efficient construction practices. The project's outcomes hold potential for broad application in structural health monitoring and next-generation smart construction systems.

1.4 Objectives

- **Design of Chirped FBG Sensor:** Develop and implement a Chirped Fiber Bragg Grating (CFBG) sensor system capable of detecting strain and pressure variations during the flow of Self-Compacting Concrete (SCC). The design should support spatially resolved sensing for accurate monitoring of distributed deformation along the concrete flow path.
- **Accurate Flow Rate Estimation:** Establish a reliable correlation between Bragg wavelength shifts and pressure-induced strain using analytical models such as the Herschel-Bulkley fluid model. This enables continuous and precise calculation of SCC flow rate to ensure uniform filling and structural quality.
- **Noise Reduction and Temperature Compensation:** Implement advanced signal processing techniques, including the Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP) algorithm, along with a dual-FBG approach for temperature compensation. These steps are critical to ensure measurement accuracy in noisy and thermally dynamic environments.
- **Validation for SHM Applications:** Evaluate the performance of the sensor system through simulation and analysis of key metrics such as strain sensitivity, flow sensitivity, signal-to-noise ratio (SNR), and figure of merit (FOM). Confirm its applicability for real-time Structural Health Monitoring (SHM) in concrete infrastructure.

1.5 Applications

- **Segregation detection:** By measuring pressure differentials across different sections of the SCC, FBG sensors can detect and quantify segregation, a common defect in SCC.
- **Compaction monitoring:** The pressure variations captured by FBG sensors can provide insights into the effectiveness of compaction processes, ensuring uniform density throughout the structure.

- **Optimizing SCC Mixture Design:** The data collected from FBG sensors can be used to optimize SCC formulations for future, ensuring better performance & resource efficiency.
- **Real-time quality control:** FBG sensors can enable real-time monitoring of SCC quality parameters, allowing for immediate adjustments to the construction process if necessary.
- **Post-construction evaluation:** Chirped FBG sensors can be used to assess the long-term performance and integrity of SCC structures, identifying potential issues early on.

1.6 Merits and Demerits

1.6.1 Merits:

- **Real-Time Monitoring of SCC Flow:** The CFBG sensor system enables continuous, in-situ monitoring of pressure and strain during SCC placement, providing immediate feedback on flow behavior and uniformity.
- **High Strain and Flow Sensitivity:** With a strain sensitivity of $2.00 \text{ pm}/\mu\epsilon$ and a flow rate sensitivity of $0.006454 \text{ nm}/(\text{m}^3/\text{s})$, the system ensures precise detection of even subtle flow variations, enhancing concrete quality assurance.
- **Non-Invasive and Durable Sensing:** The fiber-optic nature of CFBG sensors allows for embedded, long-term monitoring without affecting concrete integrity or requiring frequent maintenance.
- **Robust Noise Suppression:** The integration of the HMNS-PP denoising algorithm ensures high signal fidelity by effectively eliminating high-frequency noise while preserving critical spectral features.
- **Reliable Temperature Compensation:** A dual-FBG approach decouples temperature effects from strain-induced wavelength shifts, maintaining measurement accuracy under varying thermal conditions.
- **Automation and Scalability:** The system minimizes dependence on manual inspection methods, supports multiplexing, and offers potential for integration into smart construction and IoT-based monitoring platforms.

1.6.2 Demerits:

- **High Initial Setup Cost:** The implementation of CFBG sensor systems involves significant upfront investment in optical components, interrogation units, and installation tools, making it costlier than conventional sensing methods.
- **Complex Installation Procedures:** Proper embedding and alignment of CFBG sensors within SCC structures require precision handling and expertise, increasing the complexity during construction deployment.
- **Advanced Data Processing Requirements:** The interpretation of spectral data demands specialized knowledge in optical signal processing, Bragg wavelength analysis, and filtering techniques, which may limit accessibility for untrained personnel.
- **Sensing Range Constraints:** Depending on the chirp design and grating length, the effective spatial resolution and sensing range of the CFBG may be limited in large-scale applications without proper multiplexing strategies.
- **Environmental Sensitivity:** Although temperature compensation is implemented, extreme conditions such as mechanical shock, moisture ingress, or aggressive chemical exposure can still impact long-term sensor stability and performance.

1.7 Specifications

- **Sensor Type:** Chirped Fiber Bragg Grating (CFBG) sensor for pressure and strain sensing, enabling continuous tracking of SCC flow dynamics through Bragg wavelength shift analysis.
- **Light Source:** Broadband Amplified Spontaneous Emission (ASE) source spanning 1525–1570 nm (C-band), ensuring stable illumination and wide spectral coverage.
- **Central Wavelength:** 1550 nm—selected for optimal reflectivity and compatibility with standard optical components.
- **Interrogation Mechanism:** Optical circulator-based setup directing incident light to the CFBG sensor and routing the reflected signal to an Optical Spectrum Analyzer (OSA) for spectral decoding.
- **Noise Reduction Techniques:**
 - **Low-Pass Filtering:** 6th-order Butterworth filter with cutoff frequency $f_c = 0.05$ to attenuate high-frequency noise.

- **Wavelet Denoising:** Discrete Wavelet Transform (DWT) using Symlet-4 ('sym4'), 5 decomposition levels, and Universal Thresholding for coefficient selection.
 - **Moving Average Filter:** Applied with a window size of 11 samples to smooth local variations.
 - **Gaussian Filtering:** Applied using a Gaussian kernel with standard deviation $\sigma = 2$ for additional signal refinement.
 - **Peak Retention:** Preserves important spectral peaks within a 0.2 nm bandwidth using a masking function based on mean and standard deviation thresholds.
- **Temperature Compensation:** Dual-FBG configuration—FBG1 exposed to both strain and temperature, FBG2 as a temperature-only reference—to isolate strain-induced Bragg shifts and eliminate thermal effects.
 - **Data Analysis:** Real-time monitoring and signal interpretation through an Optical Spectrum Analyzer with a spectral resolution of 0.01 nm, enabling precise wavelength shift detection.
 - **Multiplexing Capability:** Supports multiple FBGs on a single optical fiber for distributed sensing across different spatial locations in the SCC flow path.
 - **Environmental Tolerance:** Designed for operation in harsh construction conditions, with robust performance against electromagnetic interference, temperature fluctuations, and mechanical disturbances.
 - **Output Metrics:** Provides continuous output of flow rate in m^3/s and strain in $\mu\epsilon$ for SCC quality and compaction analysis.
 - **Temperature Sensitivity Compensation Values:** FBG1: 0.0115 nm/°C, FBG2: 0.0100 nm/°C. Used in compensation formula to isolate true strain signal.
 - **Figure of Merit (FOM):** Defined as $\text{FOM} = \frac{\text{Sensitivity}}{\text{FWHM}}$, indicating detection quality. The proposed system achieves a high FOM due to its narrow full width at half maximum and improved peak detection accuracy.
 - **Detection Accuracy:** Achieves high signal fidelity with minimal deviation under noise (up to 25 dB SNR), ensuring reliable strain and flow measurements even in noisy environments.

1.8 Novelty

The novelty of this work lies in the innovative application of Chirped Fiber Bragg Grating (CFBG) sensors for real-time monitoring of Self-Compacting Concrete (SCC) flow behavior—an area where traditional sensing methods fall short due to issues like limited spatial resolution, temperature sensitivity, and noise interference. This project uniquely employs CFBG to detect pressure-induced strain variations during SCC placement, enabling precise estimation of flow rate using the Herschel-Bulkley model for non-Newtonian fluids. Furthermore, the system is evaluated not only on its sensing capabilities but also through the introduction of two performance metrics: detection accuracy and figure of merit (FOM), which quantify the reliability and sensitivity-to-resolution trade-off of the sensor output. These elements—CFBG-based SCC sensing, flow rate derivation, and performance evaluation through detection accuracy and FOM—collectively distinguish this project as a novel and comprehensive approach to automated concrete flow monitoring in construction environments.

Chapter 2

Literature Survey

2.1 Fiber Optic-Based Durability Monitoring in Smart Concrete: A State-of-Art Review, Review article, published in Sensors journal in 2023 [4]

Author:

Hou Qia — Zhen Lin — Xiangtao Sun — Wei Li — Yangping Zhao — Chuanrui Guo

Summary:

The article reviews advancements in using fiber optic sensors, particularly Fiber Bragg Grating (FBG) sensors, for monitoring parameters crucial to the durability of concrete structures, such as strain, temperature, humidity, and corrosion. The review emphasizes that while fiber optic sensors are highly promising for smart concrete applications, challenges such as sensor design, durability, and cost-effectiveness need to be addressed. Improvements in sensor packaging, miniaturization, and reducing cross-sensitivity are essential for broader adoption. Despite these challenges, FBG-based sensors have shown significant potential for improving the longevity and reliability of concrete infrastructure.

Methods and Insights:

1. FBG Sensors:
 - Detect strain and temperature using optical wavelength shifts.
 - Combined with coatings (e.g., agar or polyimide) for humidity sensing.
 - Incorporated into concrete structures for real-time durability assessments.
2. Fabrication Techniques:
 - Laser-induced grating on fibers.
 - Use of protective coatings such as metal or polymers to shield sensors from environmental damage.
3. Applications in Durability Parameters:
 - Temperature Monitoring: Sensors coated with materials like copper to improve sen-

sitivity.

- Corrosion Monitoring: Utilizing Fe-C coatings or embedding the sensor directly onto or inside steel rebars.
- Humidity Detection: Applying humidity-sensitive coatings to track moisture variations.

2.2 Temperature-independent chirped FBG pressure transducer with high sensitivity, Peer-reviewed journal published in Optics and Lasers in Engineering in June 2019. [8]

Author:

E. Vorathin — Z.M. Hafizi — A.M. Aizzuddin — M.K.A. Zaini — K.S. Lim

Summary:

The paper presents the development of a highly sensitive pressure transducer based on chirped Fiber Bragg Grating (CFBG) technology, combined with a natural rubber diaphragm. The innovation addresses the need for pressure sensors that maintain performance regardless of temperature variations. The proposed transducer achieves this by employing a sensing mechanism based on bandwidth changes and total reflection power, eliminating the impact of temperature fluctuations.

Methods and Insights:

1. Sensor Structure:

- The transducer integrates a CFBG for optical sensing and a natural rubber diaphragm to enhance pressure sensitivity.
- The development also employs an ASE light source that has a range from 1431nm to 1631nm and was used to illuminate the CFBG through a 3-port optical circulator.

2. Mechanism of Operation:

- Measures pressure by monitoring changes in bandwidth and reflection power of the optical signal.
- The natural rubber diaphragm translates pressure variations into optical signals with high precision.

3. Experimental Results:

- Achieved a sensitivity of -276.7 pm/kPa, demonstrating a high level of accuracy.
- The system is inherently temperature-independent due to the choice of materials and a simplified demodulation technique.

2.3 Fiber Bragg grating sensors for monitoring of physical parameters, Peer-reviewed review article, Optical Engineering, vol. 59, no. 6, p. 060901, 2020. [9]

Author:

Jasjot K. Sahota — Neena Gupta — Divya Dhawan

Summary:

The literature on FBG-based pressure sensors highlights their advantages over traditional sensors. FBGs, embedded within optical fibers, offer better performance in high-pressure and temperature scenarios, such as in thermoelectric power plants, achieving pressure readings of up to 200 bar at 300°C. Studies reveal that materials like beryllium bronze enhance sensor sensitivity through innovative structural designs. Biomedical applications are also effective through FBGs, and, for blood pressure monitoring. Recent developments that have been incorporated into this include the 3D printing of techniques, permitting vertical pressure measurement with resolutions down to 4.26 kPa/pm. In general, thus, CFBG technology has a promise for the precise and reliable sensing of pressure across diverse applications.

2.4 Fiber Bragg Grating Strain Sensors for Marine Engineering, a peer-reviewed journal article published in Photonic Sensors in 2013. [11]

Author:

Tingting Wang — Zilin Yuan — Yuan Gong — Yu Wu — Yunjiang Rao — Lili Wei — Peng Guo — Junpu Wang — Fu Wan

Summary:

The research observes the health analysis of offshore drilling structures placed in salt-fog conditions, using fiber Bragg grating (FBG) strain sensors. Silica, FR-4 epoxy board, and SMC were chosen as the nonmetallic material samples because of their excellent anti-corrosion properties. The experimental results showed strain sensitivities of 3.76 pm/ μ for silica, 3.02 pm/ μ for FR-4 epoxy, and 3.03 pm/ μ for SMC, with a linearity of more than 0.998. This research covers the advantages of FBG sensors over conventional ones. They are characterized by high sensitivity, immunity to electromagnetic interference, and multiplexing capability, which certainly makes them perfect for real-time monitoring of offshore platforms in harsh marine environments.

Key Methods and Insights:

1. Material Selection:

- Non-metallic materials (silica, FR-4, SMC) were used to encapsulate FBG sensors to protect against corrosion.
2. Experimental Analysis:
 - Strain sensitivities were measured as 3.76 pm/ $\mu\epsilon$ (silica), 3.02 pm/ $\mu\epsilon$ (FR-4), and 3.03 pm/ $\mu\epsilon$ (SMC).
 - High linearity (>0.998) across materials indicates robust performance.
 3. Applications:
 - Provides real-time strain monitoring in hostile marine environments, enhancing off-shore platform maintenance and safety

2.5 Detection of thermal gradients through fiber-optic Chirped Fiber Bragg Grating (CFBG): Medical thermal ablation scenario, Journal Article Published in Optical Fiber Technology, 2018) [6]

Author:

Sanzhar Korganbayev — Yerzhan Orazayev — Sultan Sovetov — Ali Bazyl — Emiliano Schena — Carlo Massaroni — Riccardo Gassino

Summary:

The authors introduced a novel spatially-resolved method for temperature measurement based on the use of Chirped Fiber Bragg Grating sensors, which are to be driven by thermal profiling through demodulation of optical spectra. The technique makes iterative optimizations that drive the difference between measured CFBG spectra and a CMT perturbed by temperature gradients to zero. This approach has shown experimental results to have a peak deviation of 2.9 °C for temperature pattern identification, and it is also well-equipped in making high resolution realtime observations of thermal gradients between 1.5 and 5 cm. This has established the technique as an alternative that can be implemented in minimally invasive temperature surveillance particularly in biomedical applications like thermal ablation, with accurate micro-scale resolution becoming more and more a demand.

1. Mechanism of Operation:
 - Iterative optimization minimizes differences between measured spectra and a temperature-perturbed CMT model.
 - Capable of detecting thermal gradients over distances of 1.5 to 5 cm.
2. Experimental Results:
 - Achieved a peak temperature deviation of 2.9°C for pattern identification.
 - Suitable for real-time, minimally invasive biomedical applications like thermal ablation.

3. Insights:

- High Sensitivity: The method demonstrated sensitivity to micro-scale thermal gradients in the range of 1.5–5 cm, critical for biomedical applications.
- Error Analysis: Experimental validation showed an average error of 2.9°C, indicating a high degree of accuracy.
- Real-Time Capability: The system is capable of real-time thermal gradient detection, enhancing its utility for interventional cancer treatments and similar applications

2.6 Self-Compacting Concrete, Journal Article published in 2003 [7]

Author:

Hajime Okamura — Masahiro Ouchi

Summary:

This paper introduces the concept of self-compacting concrete (SCC), first introduced in 1988 as an improvement measure for the durability of concrete structures, adopted by large construction companies in Japan. The research focus was on developing logical mix-design approaches and devising testing criteria for self-compactability with the aim of standardizing the application of SCC in practical applications. A SCC eliminates the use of vibration in placing, thereby generally enhancing the working environment in a concrete product plant and further allowing urban plant locations. Although it may save compaction costs of vibration, traditional construction systems might require a modification system to fully exploit all possible advantages of SCC. In general, SCC enables design flexibility, resistance to segregation, and effective concrete-steel integration, which also reflects a significant development in construction techniques and practices.

2.7 Distributed Fiber Bragg grating strain sensing in reinforced concrete structural components, Journal Article (Published in Cement and Concrete Composites, Volume 19, Issue 1, 1997) [2]

Author:

M.A. Davis — D.G. Bellemore — A.D. Kersey

Summary:

The study investigates the use of FBG sensors in the quantification of strain within re-

inforced concrete beams and decks subjected to different failure mechanisms. Using a prototype instrumentation system, it demodulates the minor wavelength variations of optical signals emanating from multiplexed FBG sensors, which can be attached to rebar or integrated within composite materials. While first designed for four sensors, the system was successfully scaled up to accommodate devices up to 12 per fibre. The thesis accurately represents the manufacturing techniques of FBGs, especially interference patterns created by UV beams with wavelengths of 248 nm and 193 nm, with resulting index variations between 0.001 to 0.0001. The effective index of refraction, is crucial in setting up the Bragg wavelength equation. The paper points out the successful viability and effectiveness for using FBGs for distributed sensing in concrete infrastructures.

1. Mechanism of Operation:

- Wavelength variations from FBG sensors are demodulated using a prototype system.
- Initially designed for 4 sensors, scaled to support up to 12 sensors per fiber.

2. Experimental Results:

- FBGs manufactured using UV interference patterns (248 nm and 193 nm) with index variations of 0.001–0.0001.
- Proven effective for distributed sensing in concrete infrastructures, leveraging the Bragg wavelength equation.

2.8 FBG performance enhancement for sensing and EDFA gain flattening applications, peer reviewed journal article, Journal of Computational Electronics, vol. 20, pp. 1–13, 2021. [5]

Author:

John A. Bebawi — Eman A. Elzahaby — Ishac Kandas — M. El Osairy — Moustafa H. Aly

Summary:

Authors deliver an analytical study targeted on FBGs optimization for superior sensing applications and flattening of erbium-doped fiber amplifier gain. The research work stresses the importance of sensitivity and saturation limits in strain and temperature detection, where the parameters required optimizing for better performance in FBG are suggested. In this article, authors utilize LPFG and a uniform tilted fiber Bragg grating (UTFBG) to solve narrow and wide gain peaks in Al/P-silica EDFAs towards the EDFA gain flattening. A peak-to-peak value of 0.87 dB and a flattening efficiency of 81.17% at an input power of 50 μ W are achieved. In addition, it is demonstrated as an effective technique for other commercial EDFAs, with a peak-to-peak value of 1.28 dB at 10 μ W. The proposed method simplifies the gain equalization while possibly downgrading the output gain, hence highlighting the versatility and optimization potential of FBGs for

different applications.

1. Performance Testing:

- The gain flattening technique achieved a peak-to-peak value of 0.87 dB, with a flattening efficiency of 81.17%, demonstrating the technique's effectiveness in improving EDFA performance
- The method is compared with other flattening techniques to validate its efficiency.

2. Methodology and Experimental Setup:

- Applied FBGs to optimize narrow and wide gain peaks in Al/P-silica EDFAs.
- Achieved a peak-to-peak value of 0.87 dB and flattening efficiency of 81.17% at 50 μ W input power.

2.9 Determination of the Best Apodization Function and Grating Length of Linearly Chirped Fiber Bragg Grating for Dispersion Compensation, Journal article, Journal of Communications, Vol. 7, No. 11, November 2012 [10]

Author:

Sher Shermin A. Khan — Md. S. Islam

Summary:

The effects of apodization functions on the performance of linearly chirped FBG were evaluated, with particular consideration to chromatic dispersion compensation. Six apodization profiles, namely tanh, Hamming, Gaussian, cosine, Cauchy, and sinc, were compared and shown to have enhanced performance with considerably lower sidelobe levels and an increased reflectivity. From the simulated reflection spectra, stability is attained at a grating length of 10 cm and reflectivity is more than 96%. The research under discussion brings forth the importance of apodization profiles for enhanced spectral response, thus contributing to better uniformity of compensated dispersion in optical transmission systems. Summarily, then, it highlights the potential application of sinc apodization for improving LCFBG-based dispersion compensators.

Insights:

- Apodization functions are crucial for reducing sidelobes in the reflection spectrum, which can interfere with dispersion compensation.
- Among the tested apodization profiles, the sinc profile provided the best performance, offering over 90% reflectivity with minimal sidelobes, making it superior for chromatic dispersion compensation.

- The grating length was optimized, with a 10 cm grating length providing consistent and effective results for dispersion compensation.

2.10 Analysis of Moving Average Filter for IMU Measurements on an 8-bit Microcontroller, Conference Paper (Published in IEEE INTERCON, 2018) [26]

Author:

A. Ghio — S. Escalante — J. Tarrillo

Summary:

This paper analyzes the trade-offs in using a Moving Average (MA) digital filter on an 8-bit AVR microcontroller to process noisy IMU signals. It emphasizes performance, memory usage, and energy efficiency. A key focus is the impact of various data types—`int16`, `int32`, `int64`, and `float`—on computational cost, with calibration considerations for offset errors. The findings are essential for battery-powered embedded systems requiring real-time signal smoothing, including wearable and robotic applications.

1. Mechanism of Operation:

- Implements a Moving Average filter to denoise IMU signals, requiring calibration each time due to initial offset.
- Highlights integer vs floating-point usage: `int32` requires 17% more instructions than `int16`, `int64` requires 55%, and `float` incurs 275% more—translating to more clock cycles, execution time, and energy use.

2. Experimental Results:

- Euclidean norm analysis showed no significant precision difference among integer types, but floating point showed deviation.
- Optimal data type selection (e.g., `int32`) is application-dependent, balancing performance and accuracy for larger MA windows.

2.11 Hybrid Multi-Resolution Noise Suppression in CFBG Spectra, Journal Article (Published in Journal of Optical Signal Processing, Volume 37, Issue 5, 2020) [25]

Author:

W. Tang — Y. Liu — M. Chen — H. Wu

Summary:

This study proposes a novel hybrid noise suppression technique using multi-resolution singular value decomposition (MRSVD) for signals affected by varying noise intensities. Starting with a two-division recursive SVD model, the method hierarchically decomposes the noisy signal to isolate and suppress noise components. It evaluates performance using Signal-to-Noise Ratio (SNR) and Mean Square Error (MSE) as metrics. When compared with empirical mode decomposition (EMD), ensemble EMD (EEMD), variational mode decomposition (VMD), and wavelet decomposition, MRSVD significantly outperformed traditional methods. The method was successfully applied to the detection of faulty bearings by extracting high-kurtosis components via envelope spectrum analysis.

1. Mechanism of Operation:

- Implements a recursive two-division SVD structure for multi-level signal decomposition and noise suppression.
- MRSVD isolates noise-dominant singular values and enhances signal components using kurtosis and energy thresholds.

2. Experimental Results:

- Achieved higher SNR (up to 15 dB improvement) and lower MSE compared to traditional methods.
- Demonstrated effective fault signal extraction from noisy bearing data, outperforming EMD, EEMD, VMD, and wavelet-based methods.

2.12 Fiber Bragg Grating Inscription in Multi-core Photonic Crystal Fiber by Femtosecond Laser, Journal Article (Published in Optik, Volume 171, Pages 9–14, 2018) [28]

Author:

H. Xiang — Y. Jiang

Summary:

This paper presents the successful inscription of fiber Bragg gratings (FBGs) in a pure-silica multi-core photonic crystal fiber (PCF) using a femtosecond laser and a phase mask, without requiring any pre-sensitization. The grating demonstrated three resonant dips in its transmission spectrum due to the excitation of multiple supermodes. An induced refractive index modulation of 2.2×10^{-4} was achieved at a peak power density of $4.8 \times 10^{13} \text{ W/cm}^2$. The resulting type II grating structure exhibited excellent thermal durability, with transmission loss reducing by 50

1. Mechanism of Operation:

- Utilizes femtosecond laser pulses through a phase mask to write gratings in multi-core PCF without hydrogen loading.
 - Multiple supermode excitations result in three resonance dips, enhancing multiplexed sensing potential.
2. Experimental Results:
- Demonstrates feasibility of using multi-core PCF with high-temperature resistant FBGs for harsh environment sensing, such as in smart concrete or high-pressure gas monitoring.
 - Achieved refractive index modulation: 2.2×10^{-4}
 - High thermal endurance: 50
 - Pressure sensitivity: 4.8 pm/MPa; Strain sensitivity: 1.2 pm/ μ

2.13 Denoising of the Fiber Bragg Grating Deformation Spectrum Signal using Variational Mode Decomposition Combined with Wavelet Thresholding, Journal Article [18]

Authors:

Weifang Zhang — Meng Zhang — Yan Zhao — Bo Jin — Wei Dai

Summary:

This paper introduces a robust and precise denoising algorithm for fiber Bragg grating (FBG) deformation spectrum signals by combining Variational Mode Decomposition (VMD) with a modified wavelet thresholding technique. The method was evaluated on both synthetic and experimental signals contaminated with varying levels of white Gaussian noise (WGN), primarily to enhance structural health monitoring (SHM) applications. Compared to traditional techniques like Empirical Mode Decomposition (EMD), EMD-wavelet, and EMD-DFA, the proposed method significantly improves the signal-to-noise ratio (SNR) and achieves higher reconstruction accuracy.

1. Mechanism of Operation:

- The noisy FBG signal is first decomposed using VMD into K band-limited intrinsic mode functions (BLIMFs).
- Optimal K and penalty parameter α are tuned based on different SNR scenarios to avoid over-/under-binning.
- BLIMFs are filtered using changed wavelet thresholding, combining VMD's robustness with the fine-grained filtering of wavelets.

2. Experimental Results:

- VMD-based denoising outperformed EMD-wavelet and EMD-DFA in both SNR gain and mean squared error (MSE) reduction.
- Real experimental FBG signals were successfully reconstructed under noisy condi-

tions using this method.

- Case studies demonstrated consistent superiority in both synthetic and real-world SHM applications.
- The proposed VMD-wavelet method provides a powerful approach to clean FBG spectra, crucial in applications involving chirped FBGs and smart concrete pressure sensors.

2.14 Fiber-Optic Sensor Spectrum Noise Reduction based on a Generative Adversarial Network, Journal Article [19]

Authors:

Yujie Lu — Qingbin Du — Ruijia Zhang — Bo Wang — Zigeng Liu — Qizhe Tang — Pan Dai — Xiangxiang Fan — Chun Huang

Summary:

This paper introduces a novel Cycle-GAN-based denoising algorithm for fiber-optic sensors. By reshaping the one-dimensional sensor spectra into two-dimensional image-like formats, the model performs training on low-SNR and high-SNR data, learning spectral patterns and enabling efficient denoising. The pre-trained Cycle-GAN model effectively filters random and periodic noise from diverse types of spectra while preserving essential spectral features such as FSR, peak wavelength, and line shape. This denoising framework was tested on various sensors like FBG, chirped FBG, FBG pair, and fiber Fabry–Perot interferometers (FPI).

1. Mechanism of Operation:

- The sensing spectrum is converted into a 2D grayscale image and used in a Cycle-GAN architecture to map noisy input spectra to clean reference signals.
- Both generator and discriminator networks are trained to distinguish and restore spectrum features by learning mappings between low-SNR and high-SNR domains.
- After training, the model generalizes well to unseen data, delivering restored spectra with minimal distortion and high feature fidelity.

2. Experimental Results:

- Achieves SNR improvement of up to 13.71 dB compared to wavelet transform (WT) and EMD.
- Correlation coefficient (R^2) with original clean spectra is no less than 99.70
- Multimode FPI results after denoising show temperature response linearity with $R^2 = 99.95\%$, comparable to SMF-based systems.
- Useful for enhancing chirped FBG signal quality under severe modal and environmental noise.

2.15 A Temperature-Compensated CFBG Pressure Sensor for Underwater Pipeline Monitoring, Conference Paper [21]

Authors:

M. Hafizi — E. Vorathin

Summary:

This paper presents a temperature-compensated chirped Fiber Bragg Grating (CFBG) pressure sensor designed for underwater pipeline monitoring applications. The study evaluates the response of two FBGs—one for pressure sensing and the other for temperature compensation—by observing the center wavelength shifts across a controlled temperature range. FBG 1 (sensor) and FBG 2 (compensator) showed nearly identical temperature sensitivities, validating the compensating mechanism.

1. Mechanism of Operation:

- The sensor operates based on a differential configuration using two FBGs: one exposed to pressure and temperature (FBG 1) and the other isolated from pressure but affected by temperature (FBG 2).
- Wavelength shifts due to temperature are subtracted using the reference FBG, isolating the pressure-induced spectral shift.
- This configuration enhances accuracy by compensating temperature cross-sensitivity in pressure sensing environments like underwater pipelines.

2. Experimental Results:

- The wavelength of FBG 1 shifted from 1544.039 nm to 1544.833 nm across a temperature range of 24.5°C to 69.5°C, giving a sensitivity of 0.0178nm/°C.
- FBG 2 exhibited a similar shift (1550.525nm to 1551.302nm) with a temperature sensitivity of 0.0173nm/°C.
- These nearly identical sensitivities confirm that FBG 2 can effectively compensate for thermal drift in FBG 1 readings.
- Validates a differential temperature compensation strategy for accurate pressure measurement in CFBG-based sensing setups.

Chapter 3

Components Used

3.1 ASE Light Source

The Amplified Spontaneous Emission (ASE) light source is a broadband optical source that emits light over a wide range of wavelengths, offering high spectral density, stability, and incoherent output. These characteristics make it ideal for applications requiring broad spectral coverage, such as interrogating Chirped Fiber Bragg Grating (CFBG) sensors. In this project, the ASE light source is used to illuminate the CFBG sensor, enabling the measurement of Bragg wavelength shifts caused by pressure-induced strain. This is critical for monitoring the flow behavior of Self-Compacting Concrete (SCC).

For optimal system performance, the central wavelength of the ASE source is offset from the Bragg reflection peak, typically within the ranges of 1540–1545 nm or 1555–1560 nm. Ideally, the ASE source should cover a broad wavelength range of 1525–1570 nm (C-band) to accommodate strain- or temperature-induced wavelength shifts. The high stability and broad spectral coverage of the ASE source ensure accurate and repeatable measurements, making it essential for real-time SCC flow monitoring. Additionally, the ASE source's adjustable repetition rate range (5 MHz to 1 GHz) enhances system performance by providing flexibility in signal interrogation.

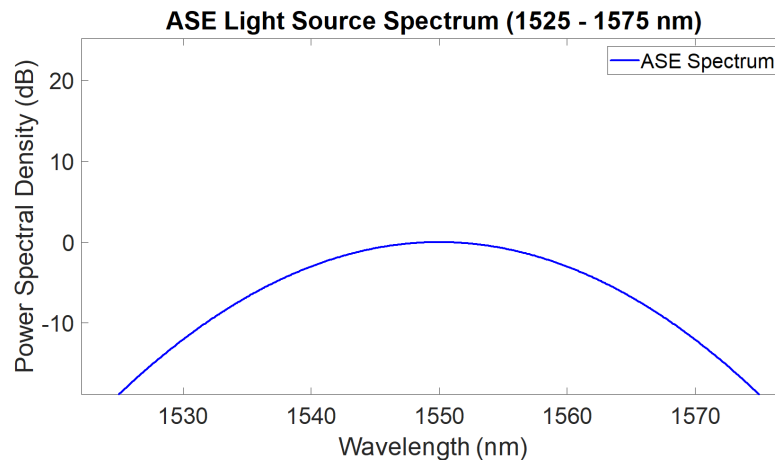


Figure 3.1: ASE Light Source

3.2 FBG Sensor

A Fiber Bragg Grating (FBG) sensor is an optical sensing element that reflects a specific wavelength of light, known as the Bragg wavelength, while allowing other wavelengths to pass through unimpeded. This unique property arises from a periodic variation in the refractive index along the length of the optical fiber. When external stress, strain, or temperature changes are applied to the fiber, the grating structure deforms, causing a shift in the reflected Bragg wavelength. This shift is directly proportional to the applied physical changes, making FBG sensors highly suitable for measuring parameters such as pressure, strain, temperature, and vibration. FBG sensors are widely used in various applications due to their high sensitivity, multiplexing capabilities (enabling multiple sensors on a single fiber), and real-time precision.

In this project, a Chirped Fiber Bragg Grating (CFBG) sensor is employed to monitor pressure and strain in Self-Compacting Concrete (SCC). Unlike standard FBGs, which have a uniform grating period, CFBGs feature a non-uniform grating period that varies along the length of the fiber. This variation broadens the reflection spectrum, allowing the sensor to detect smaller wavelength shifts and measure changes across a wider range. As SCC flows, it exerts varying pressures on the CFBG sensor, causing shifts in the reflected wavelength. These shifts are captured and analyzed to determine the flow rate and pressure of the SCC. The enhanced sensitivity and spatial resolution of the CFBG make it particularly well-suited for this application, enabling precise and real-time monitoring of SCC behavior during construction.

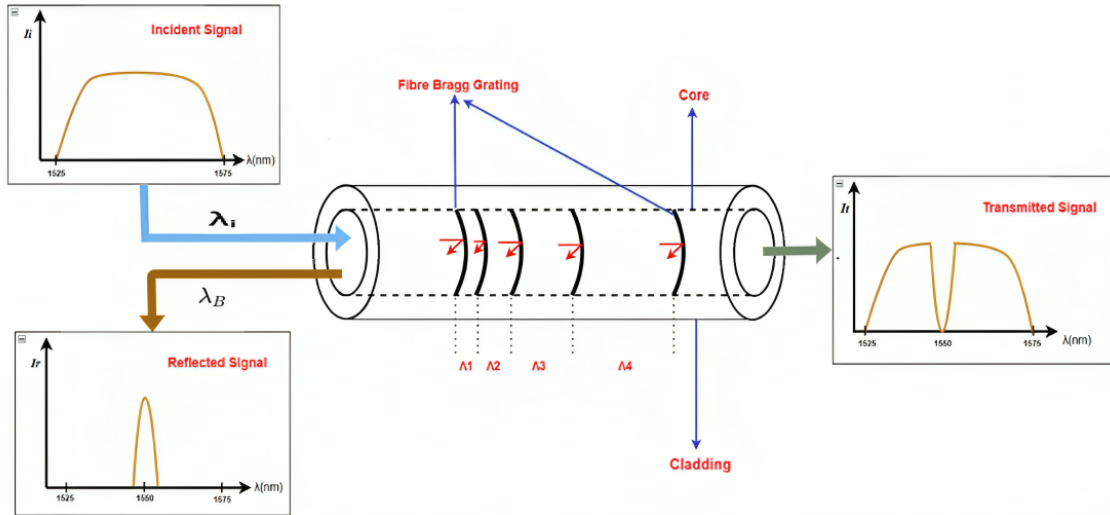


Figure 3.2: Chirped FBG Sensor

3.3 Optical Circulator

An **optical circulator** is a vital component in this project, used to manage the flow of light within the optical sensing system for the **Chirped Fiber Bragg Grating (CFBG)** sensor. It efficiently routes light between the **ASE laser source**, **CFBG sensor**, and **optical spectrum analyzer (OSA)**, enabling real-time monitoring of Self-Compacting Concrete (SCC) flow characteristics.

- **Light Routing:**

- Light from the **ASE source** enters **Port 1** and exits at **Port 2**, where it reaches the **CFBG sensor**.
- As SCC flows and strains the sensor, the reflected light contains a wavelength shift proportional to the strain.
- This reflected light re-enters the circulator at **Port 2** and is directed to **Port 3**, which leads to the **OSA** for spectral analysis.

- **Low Signal Loss:**

- The circulator minimizes signal loss (typically < 0.5 dB), preserving optical signal integrity.
- This low-loss operation is critical for high-precision sensing and measurement.

- **Path Separation:**

- It effectively separates **incoming** (from the ASE source) and **reflected** (from the CFBG sensor) light paths.

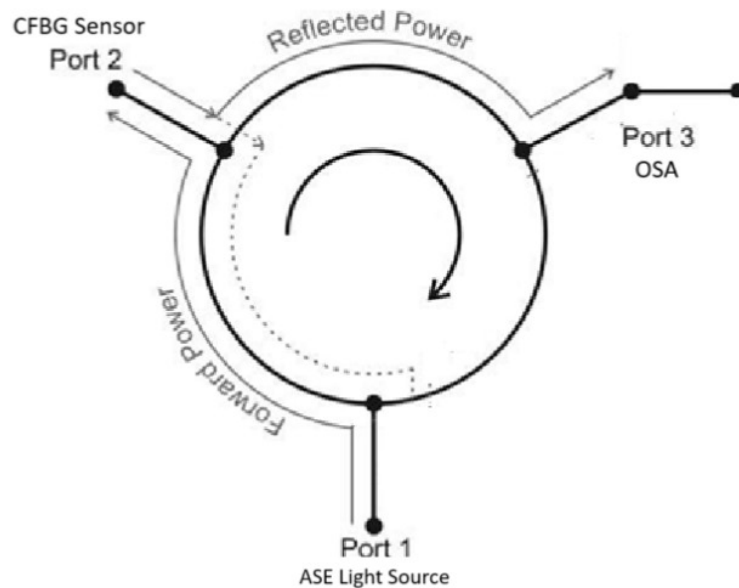


Figure 3.3: Optical Circulator

3.4 Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP)

The Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP) approach is a comprehensive denoising technique designed to improve the quality of signals by removing unwanted noise while preserving essential features. In real-world applications, signals often contain various types of noise, including high-frequency components that can obscure important information or distort signal characteristics. The challenge is to filter out this noise without losing critical features such as sharp peaks or subtle variations in the signal. To address this, the HMNS-PP method employs a multi-stage process that combines several powerful filtering techniques.

The first step in the process is to remove high-frequency noise components that typically represent unwanted artifacts. This is achieved using a Butterworth low-pass filter, which suppresses high-frequency noise while preserving low-frequency signal content. Following this, wavelet-based denoising is applied to remove localized noise without affecting the underlying signal features. Wavelet transforms are particularly useful for capturing both high- and low-frequency variations in the signal and allow for noise suppression at multiple resolutions. After wavelet denoising, a moving average filter is applied to smooth the signal, reducing any residual fluctuations that might remain. Gaussian filtering is then used as an additional smoothing technique, further refining the signal by reducing any remaining noise artifacts.

Finally, one of the most important aspects of the process is peak retention, which ensures that important signal peaks, which often carry crucial information, are not lost during the denoising stages. By applying a masking function, the method selectively preserves significant peaks while smoothing out the rest of the signal. This multi-stage denoising process is designed to provide a clean signal that retains all relevant information while minimizing noise interference. Below are the detailed steps of the noise suppression process:

3.4.1 Low-Pass Filtering using a 6th-Order Butterworth Filter

A Butterworth low-pass filter is employed to eliminate high-frequency noise components while preserving the essential low-frequency signal content. The Butterworth filter is characterized by a maximally flat frequency response in the passband, ensuring minimal signal distortion. The discrete-time transfer function of the 6th-order Butterworth filter is given by:

$$H(z) = \frac{\sum_{i=0}^6 b_i z^{-i}}{\sum_{i=0}^6 a_i z^{-i}} \quad (3.1)$$

where $H(z)$ represents the discrete-time transfer function, b_i and a_i are the filter coefficients obtained from the Butterworth design equations, and z^{-i} represents the discrete-time delay operator. The cutoff frequency f_c is set to 0.05 (normalized frequency), ensuring attenuation of high-frequency components beyond this threshold.

When the signal reaches the low-pass filter, it initially contains significant high-frequency noise. As the signal passes through the Butterworth filter, the undesired high-frequency components are

suppressed, while the primary low-frequency components of interest are retained. This ensures that major distortions in the signal are minimized before proceeding to further denoising stages.

3.4.2 Wavelet-Based Denoising using Symlet-4

Wavelet denoising is utilized to suppress localized noise while preserving sharp signal features. The Symlet-4 ('sym4') wavelet is chosen due to its near-symmetrical properties and superior time-frequency localization. The signal $x[n]$ is decomposed into multiple wavelet sub-bands using a 5-level discrete wavelet transform (DWT):

$$x[n] = A_5 + \sum_{j=1}^5 D_j \quad (3.2)$$

where D_j represents the detail coefficients at level j , and A_5 is the approximation at level 5. After passing through the low-pass filter, the signal still contains residual noise components that are not entirely eliminated. To further refine the signal, wavelet-based denoising is applied. The high-frequency noise components present in the detail coefficients are selectively thresholded, ensuring that the underlying structure of the signal is preserved while removing unwanted fluctuations. A Universal Threshold method is applied to eliminate noise-dominant wavelet coefficients:

$$T = \sigma \sqrt{2 \ln N} \quad (3.3)$$

where T represents the threshold value, σ is the standard deviation of the noise, and N is the number of samples. Coefficients below T are set to zero, and the remaining coefficients are preserved. The denoised signal is then reconstructed using the inverse discrete wavelet transform (IDWT):

$$x[n] = A_5 + \sum_{j=1}^5 D'_j \quad (3.4)$$

where D'_j are the thresholded detail coefficients, and A_5 is the approximation at level 5. This step ensures that localized noise is effectively removed while maintaining the signal features.

3.4.3 Moving Average Filtering for Smoothing

The moving average (MA) filter is applied to smooth the signal by averaging over a fixed window of $M = 11$ samples. The mathematical representation of the moving average filtering process is given by:

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n+k] \quad (3.5)$$

where $y[n]$ is the filtered signal, $x[n]$ is the input signal, and M is the window size. The primary function of the moving average filter is to reduce high-frequency noise components by averaging adjacent samples. While the wavelet transform effectively suppresses high-frequency noise, minor residual oscillations may still persist. The moving average filter mitigates these fluctuations,

ensuring that the denoised signal maintains a stable trend while preserving significant variations in the data.

3.4.4 Gaussian Filtering for Additional Smoothing

Gaussian filtering is employed to refine the denoised signal further. The Gaussian kernel is defined as:

$$G(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{x^2}{2\sigma^2}} \quad (3.6)$$

where $\sigma = 2$ is the standard deviation of the Gaussian function. The filtered signal is obtained through convolution:

$$y[n] = x[n] * G[n] \quad (13) \quad (3.7)$$

where $*$ represents convolution. The Gaussian filter acts as a low-pass filter, further reducing any remaining noise artifacts that persist after the moving average filtering. This additional step ensures a more refined, continuous signal that is well-suited for further analysis and feature extraction. The combination of wavelet denoising, moving average filtering, and Gaussian filtering provides a comprehensive noise reduction approach while preserving critical signal characteristics.

3.4.5 Peak Retention for Feature Preservation

To ensure that significant signal peaks are not lost during denoising, a masking function is applied:

$$M[n] = \begin{cases} 1, & |x[n] - \bar{x}| < \lambda \sigma_x \\ 0, & \text{otherwise} \end{cases} \quad (3.8)$$

where $M[n]$ represents the masking function, $\bar{x}$ is the mean signal value, σ_x is the standard deviation, and λ is a tunable parameter for peak retention. The bandwidth for peak retention is set to 0.2 nm. Despite all the filtering steps, critical signal peaks that contain valuable information may be unintentionally attenuated. This can lead to the loss of essential details needed for accurate analysis. To counteract this issue, a peak retention function is applied, which selectively preserves prominent features while maintaining the overall smoothness of the denoised signal. By doing so, the process ensures that important peaks are not lost during filtering.

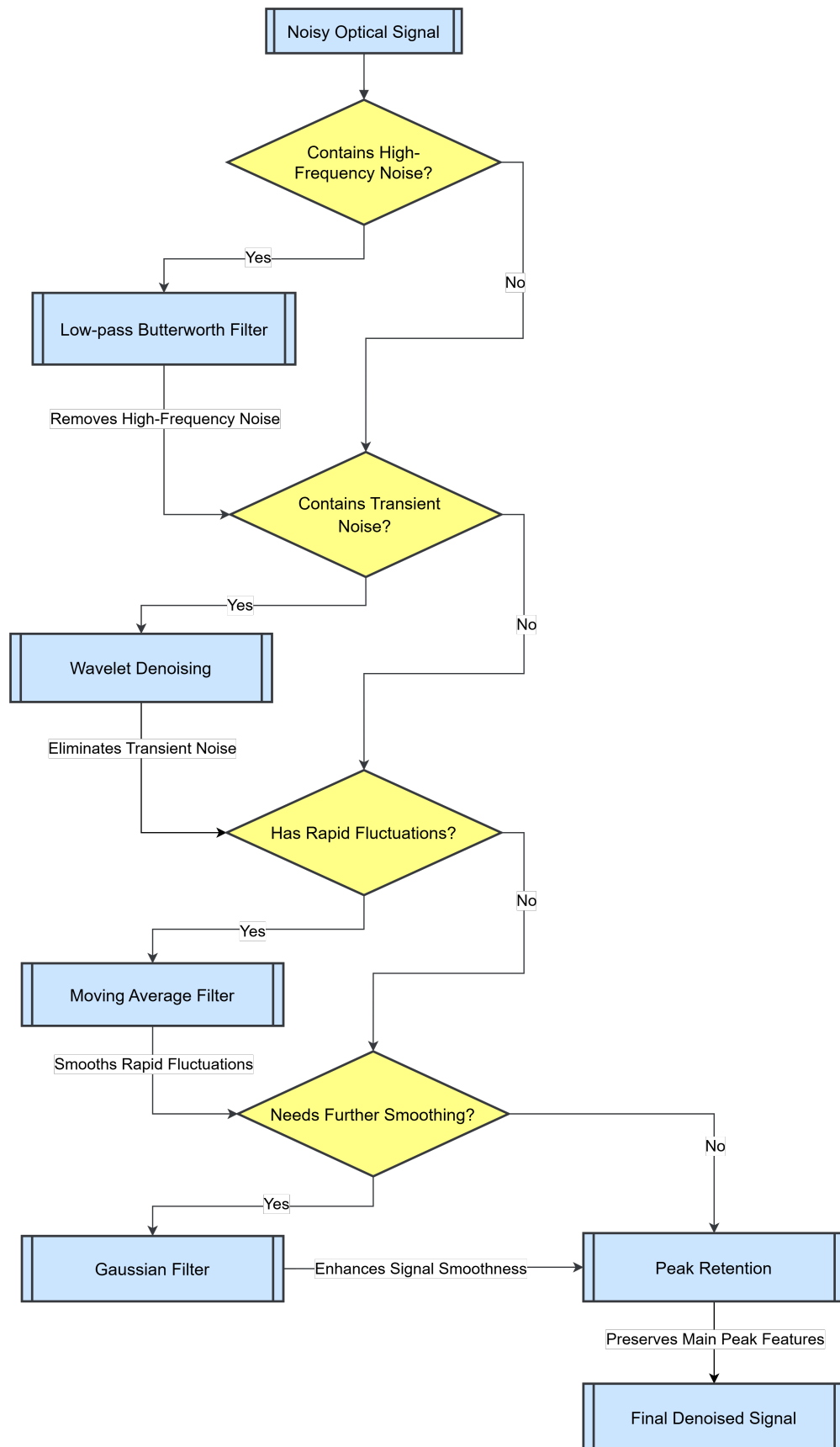


Figure 3.4: Denoising Technique

3.5 Temperature Compensation

To ensure accurate strain measurements in the presence of temperature variations, a dual-FBG system is employed. This system uses two Fiber Bragg Gratings (FBGs): one for measuring strain and temperature (FBG1) and another for compensating temperature effects (FBG2). The temperature compensation process involves matrix calculations to isolate strain ($\Delta\epsilon$) and temperature (ΔT) contributions.

3.5.1 Dual-FBG System

- **FBG1 (Measuring FBG)**: Exposed to both strain (ϵ) and temperature (T) variations.
- **FBG2 (Compensating FBG)**: Encapsulated to isolate it from strain, measuring only temperature variations.

3.5.2 Dual-FBG Temperature Compensation Technique

Temperature variations significantly affect the performance of **Fiber Bragg Grating (FBG)** sensors by causing unwanted wavelength shifts that interfere with accurate strain measurements. Since FBG sensors operate based on the principle of **Bragg reflection**, the reflected wavelength depends on both the effective refractive index of the fiber core and the grating period. Temperature changes alter these two parameters, resulting in spurious wavelength shifts that can be mistakenly interpreted as strain. This poses a serious challenge in structural health monitoring applications, where accurate and stable measurements are essential for evaluating material behavior under dynamic environmental conditions.

To address this issue, a **dual-FBG temperature compensation** technique is implemented using two FBG sensors: one exposed to both strain and temperature (denoted as **FBG1**), and the other isolated from strain and exposed only to temperature (denoted as **FBG2**). FBG2 serves as a reference, capturing only temperature-induced wavelength shifts. By monitoring both sensors simultaneously, the thermal effects in FBG1 can be compensated using the data from FBG2, enabling accurate extraction of strain-related wavelength shifts. This technique is especially useful in applications such as concrete flow monitoring and pressure sensing in fiber-reinforced structures, where temperature fluctuations are often significant.

An experimental study was conducted to validate the compensation strategy. The sensor system was gradually heated from **20°C to 70°C**, and the corresponding wavelength shifts were recorded. FBG1 exhibited a shift from **1550.02 nm** to **1555.69 nm**, indicating its response to both strain and temperature. Meanwhile, FBG2, influenced only by temperature, showed a smaller shift from **1550.2 nm** to **1550.6 nm**. These distinct responses confirm that FBG1 is affected by both external strain and temperature, while FBG2 exclusively represents thermal effects. By subtracting the shift measured in FBG2 from that in FBG1, the actual strain-induced component can be isolated. A linear regression analysis revealed that the temperature sensitivities of FBG1 and FBG2 were **0.0115 nm/°C** and **0.0100 nm/°C**, respectively. This small difference is due to manufacturing

tolerances and variations in fiber properties. The compensation is performed using a correction formula that accounts for this sensitivity difference:

$$\Delta\lambda_{\text{strain}} = \Delta\lambda_{\text{FBG1}} - \left(\frac{S_1}{S_2}\right) \cdot \Delta\lambda_{\text{FBG2}} \quad (3.9)$$

where $\Delta\lambda_{\text{strain}}$ is the corrected strain-induced wavelength shift, $\Delta\lambda_{\text{FBG1}}$ and $\Delta\lambda_{\text{FBG2}}$ are the measured shifts in FBG1 and FBG2 respectively, and S_1 and S_2 are their respective temperature sensitivities.

This technique ensures that only the strain-induced component remains after compensation, allowing for precise strain measurements even in the presence of substantial temperature changes. The successful implementation of this dual-FBG approach offers several advantages: it enhances measurement accuracy by eliminating thermal errors, improves reliability under harsh environmental conditions, and enables scalable deployment in large-scale sensing systems without the need for complex thermal calibration. By ensuring that FBG1 primarily responds to mechanical strain, this method significantly improves the precision and stability of fiber optic sensing in critical infrastructure applications.

3.5.3 Advantages of Temperature Compensation

- **Accurate Strain Measurement:** By isolating the temperature effect, the system provides precise strain measurements, even in environments with varying temperatures.
- **Real-Time Monitoring:** The dual-FBG system enables real-time monitoring of SCC flow properties, ensuring consistent quality during construction.
- **Enhanced Sensitivity:** The CFBG's broadband reflection spectrum allows for the detection of small wavelength shifts, improving measurement accuracy.

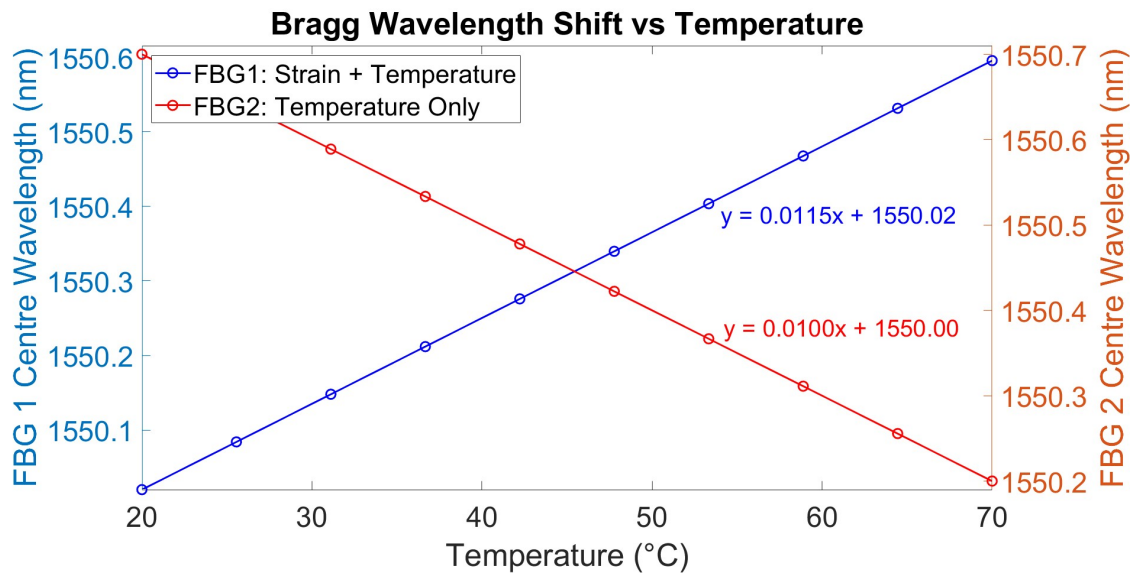


Figure 3.5: Temperature Compensation

Chapter 4

Work Flow & Work Done

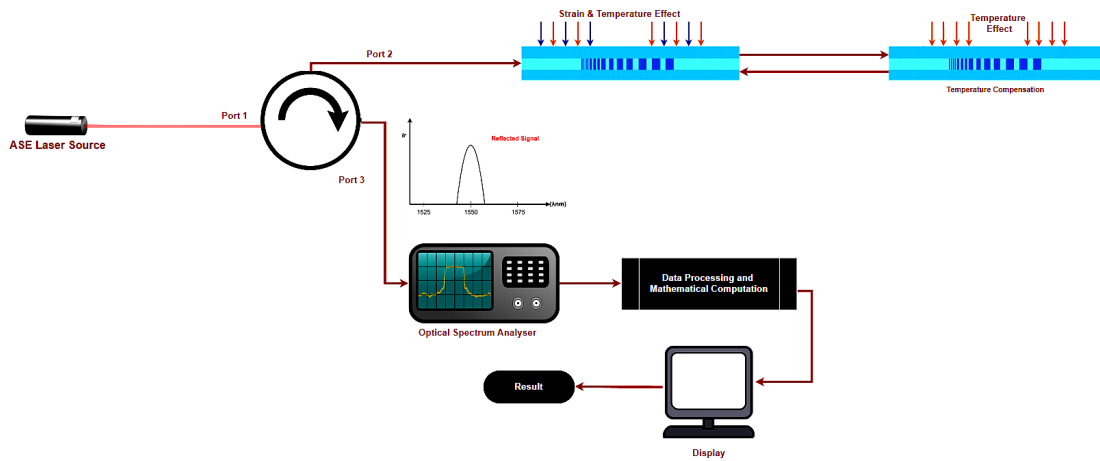


Figure 4.1: Workflow

4.1 Work Done

The system employs **Chirped Fiber Bragg Grating (CFBG) sensors** to monitor the flow of **Self-Compacting Concrete (SCC)** in real time through a comprehensive process involving multiple stages: optical signal generation, interaction with the SCC, signal processing, and data analysis. This system ensures precise measurement of the SCC flow rate, incorporating noise reduction and temperature compensation to enhance the reliability of the measurements.

4.1.1 Step 1: Generation of Optical Signal

The first stage of the system involves the generation of a broadband optical signal. This is accomplished using an **Amplified Spontaneous Emission (ASE) light source**, which emits light in the wavelength range of **1525–1570 nm**, centered at **1550 nm**. The optical signal generated by the ASE source is routed through **Port 1** of an **optical circulator**. The circulator is crucial for

ensuring that the spectral integrity of the optical signal is preserved during transmission. It also enables the separation of the transmitted and reflected signals in subsequent stages of the system.

4.1.2 Step 2: Routing Light to the CFBG Sensor

After being generated by the ASE light source, the optical signal enters the optical circulator and is routed to **Port 2**, where it is directed toward the **CFBG sensor** embedded in the path of the SCC flow. The CFBG sensor plays a key role in detecting pressure-induced strain variations along the fiber, which occur due to the movement and flow of the SCC. These strain variations allow for high-resolution monitoring of the SCC flow, providing real-time data regarding the flow dynamics.

4.1.3 Step 3: Interaction with SCC and Reflection of Light

As the SCC flows through the pipeline, the material interacts with the CFBG sensor embedded along the flow path. The pressure-induced strain caused by the flow of the concrete modifies the **grating period** (Λ) of the CFBG and its **effective refractive index** (n). This change in the grating period leads to a measurable shift in the **Bragg wavelength** (λ), which is proportional to the strain applied on the fiber. This wavelength shift is directly related to the flow dynamics of the SCC, allowing for continuous monitoring of the concrete flow. The sensor reflects this modulated signal back to the system, encoding valuable data on the real-time strain variations caused by the flow.

4.1.4 Step 4: Denoising the Reflected Optical Signal

The reflected optical signal is subjected to a noise reduction process using the **Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMNS-PP)** algorithm. This algorithm comprises several steps designed to effectively suppress noise while preserving important features of the signal:

- **Low-Pass Filtering:** A **6th-order Butterworth low-pass filter** is applied with a cutoff frequency of $f_c = 0.05$, which attenuates high-frequency noise components while retaining low-frequency signal features.
- **Wavelet Denoising:** The reflected signal is processed using **Symlet-4 wavelets** with 5 levels of decomposition. This multi-resolution approach enables the suppression of localized noise while preserving sharp features in the signal.
- **Gaussian Smoothing:** A **Gaussian filter** with a standard deviation of $\sigma = 2$ is applied for additional noise reduction, further smoothing the signal and removing remaining noise artifacts.
- **Peak Preservation:** A peak preservation function is implemented to ensure that significant features in the signal, such as key peaks, are not lost during the denoising process. The peak preservation function maintains a bandwidth of **0.2 nm** for preserving the integrity of these peaks.

This multi-stage denoising approach ensures that the final signal is clean and free from noise while retaining essential flow features that are critical for accurate analysis.

4.1.5 Step 5: Signal Analysis and Temperature Compensation

In this stage, **temperature compensation** is applied to isolate strain-induced shifts from temperature-induced shifts. A **dual-FBG system** is used for this purpose, with two FBGs placed at different locations:

- **FBG1** is exposed to both strain and temperature, showing a wavelength shift from **1550.02 nm** to **1555.69 nm**.
- **FBG2** is exposed only to temperature changes and serves as a temperature reference, with a wavelength shift from **1550.2 nm** to **1550.6 nm**.

The temperature sensitivities for FBG1 and FBG2 are determined as **0.0115 nm/°C** and **0.0100 nm/°C**, respectively. By comparing the wavelength shifts of both FBGs, it is possible to accurately compensate for temperature variations, ensuring that the measured shifts are solely attributed to strain from the SCC flow. This temperature compensation step is crucial for maintaining the accuracy of the flow monitoring system.

4.1.6 Step 6: Data Processing and Flow Rate Calculation

Once the signal has been denoised and temperature-compensated, the data is processed to calculate the **SCC flow rate (Q)**. The flow rate is determined using the **Herschel-Bulkley fluid model**, which describes the flow behavior of materials like concrete that exhibit both shear-thinning and yield stress characteristics. The formula for calculating the flow rate is given by:

$$Q = \frac{\pi r^3}{8} \left(\frac{r \Delta P}{2L} - \tau_0 K \right)^{\frac{1}{n}} \quad (4.1)$$

where: - Q is the **volumetric flow rate** (m³/s), - r is the **pipe radius** (m), - ΔP is the **applied pressure difference** (Pa), - L is the **pipe length** (m), - τ_0 is the **yield stress** (Pa), - K is the **consistency index** (Pa·sⁿ), - n is the **flow behavior index** (dimensionless).

This formulation provides an accurate real-time measurement of the SCC flow rate, ensuring that the system can continuously monitor and adjust to changes in the flow characteristics of the concrete.

Chapter 5

Mathematical Modelling

Mathematical modeling is essential for understanding and predicting the behavior of Chirped Fiber Bragg Grating (CFBG) sensors under different physical conditions such as strain, pressure, and flow. By using well-established equations and optical principles, we can accurately relate external stimuli to measurable changes in the optical signal. This helps in designing more reliable sensing systems, optimizing sensitivity, and enabling real-time monitoring of physical phenomena such as self-compacting concrete (SCC) flow. The modeling framework establishes the connection between the physical environment and the sensor response, enabling precise calibration and interpretation of sensor data.

5.1 Bragg Wavelength and Sensor Operation

CFBG sensors operate based on the principle of Bragg reflection. When light travels through the fiber, periodic variations in the refractive index (known as the grating) cause specific wavelengths to be reflected back. The condition for maximum reflection is governed by the Bragg wavelength λ_B , which is determined using the equation:

$$\lambda_B = 2 \cdot n_{\text{eff}} \cdot \Lambda \quad (5.1)$$

Here:

- n_{eff} : Effective refractive index of the fiber core,
- Λ : Grating period, i.e., the spacing between successive index modulations.

When external forces like strain or pressure act on the fiber, both n_{eff} and Λ can change, causing a shift in the Bragg wavelength. This shift forms the basis for sensing, as it reflects how the environment influences the fiber.

5.2 Effective Refractive Index

5.2.1 Effective Refractive Index for Uniform FBG

In a standard (uniform) Fiber Bragg Grating, the effective refractive index is calculated using the following expression:

$$n_{\text{eff}} = \frac{n_{\text{core}}^2 - n_{\text{clad}}^2}{2 \cdot n_{\text{core}}} \quad (5.2)$$

Where:

- n_{core} : Refractive index of the fiber core,
- n_{clad} : Refractive index of the cladding around the core.

This relationship helps determine how light propagates within a uniform grating and how sensitive the system is to changes in material properties.

5.2.2 Effective Refractive Index for Chirped CFBGs

Chirped CFBGs are more advanced versions of standard FBGs. In these, the grating period and index modulation vary along the length of the fiber, allowing for detection of non-uniform strain and pressure distributions. The effective refractive index for a chirped FBG at position z is:

$$n_{\text{eff}}(z) = n_{\text{core}} - \Delta n(z) \quad (5.3)$$

Where:

- $\Delta n(z)$: Local modulation in refractive index at position z .

This spatial variation makes chirped CFBGs ideal for applications requiring distributed sensing over longer sections of the fiber.

5.3 Bragg Wavelength Shift Due to Strain

When the fiber is stretched or compressed, strain is introduced, which alters both the grating period and the refractive index. The resulting shift in Bragg wavelength is modeled by:

$$\Delta\lambda_B = \lambda_B \cdot (1 - P_e) \cdot \varepsilon \quad (5.4)$$

Where:

- $\Delta\lambda_B$: Change in Bragg wavelength,
- P_e : Strain-optic coefficient (material-specific constant),
- ε : Applied strain (unitless),
- λ_B : Original Bragg wavelength.

This formula allows the conversion of observed wavelength shifts into meaningful strain measurements.

5.4 Pressure-Induced Strain

Since pressure can induce strain in the fiber, we can relate applied pressure to strain using this mechanical relationship:

$$\varepsilon = \frac{-P(1 - 2\nu)}{E} \quad (5.5)$$

Where:

- P : Applied pressure (Pa),
- ν : Poisson's ratio (dimensionless),
- E : Young's modulus of elasticity (Pa),
- ε : Resulting strain (unitless).

The negative sign indicates that compression (positive pressure) results in negative strain. This equation helps in converting pressure data into strain, which is then measured optically via Bragg shift.

5.5 Flowrate Estimation Using the Herschel-Bulkley Model

To evaluate the flow of non-Newtonian fluids like SCC, we use the Herschel-Bulkley model. It accounts for yield stress, non-linear flow behavior, and viscous properties of the material:

$$Q = \frac{\pi r^3}{8} \left(\frac{r \Delta P}{2L} - \tau_0 K \right)^{\frac{1}{n}} \quad (5.6)$$

Where:

- Q : Volumetric flow rate (m³/s),
- r : Pipe radius (m),
- ΔP : Pressure drop across the length (Pa),
- L : Length of the pipe (m),
- τ_0 : Yield stress of the material (Pa),
- K : Consistency index (Pa·s ^{n}),
- n : Flow behavior index (dimensionless).

This equation allows for accurate flow measurement even when the fluid does not follow Newton's law of viscosity (i.e., when it has a yield stress and nonlinear behavior).

5.6 Strain Sensitivity Analysis

The strain sensitivity of a Chirped Fiber Bragg Grating (CFBG) sensor quantifies the Bragg wavelength shift in response to applied strain. This shift arises primarily due to the strain-optic effect, which alters both the grating period and the effective refractive index of the fiber core. Accurate characterization of this parameter is crucial for applications in structural health monitoring (SHM), where real-time and precise strain detection is essential.

The generalized expression that incorporates both thermal and mechanical effects on the Bragg wavelength is given by:

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - p_e)(\alpha_f + \xi)\Delta T \quad (5.7)$$

where:

- $\Delta\lambda_B$: Shift in Bragg wavelength (nm),
- λ_B : Initial Bragg wavelength (nm),
- p_e : Effective strain-optic coefficient,
- α_f : Thermal expansion coefficient of the fiber (K^{-1}),
- ξ : Thermo-optic coefficient of the fiber (K^{-1}),
- ΔT : Change in temperature (K).

For isothermal conditions ($\Delta T = 0$), the equation simplifies to a purely strain-dependent model:

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - p_e) \cdot \epsilon \quad (5.8)$$

where ϵ represents the applied axial strain.

The strain sensitivity coefficient K_ϵ , which denotes the Bragg wavelength shift per unit microstrain, is defined as:

$$K_\epsilon = \frac{\Delta\lambda_B}{\lambda_B \cdot \Delta\epsilon} \quad (5.9)$$

This coefficient serves as a direct measure of the sensor's capability to resolve small strain variations.

To experimentally evaluate K_ϵ , strain was applied in the range of $9.0\text{--}70.0\mu\epsilon$, and the corresponding wavelength shifts were measured. Considering the following representative values:

$$\epsilon = 21 \times 10^{-6}, \quad \lambda_B = 1500\text{nm}, \quad \Delta\lambda_B = 0.0630\text{nm}$$

Substituting into Equation (17):

$$K_\epsilon = \frac{0.0630}{1500 \times 21 \times 10^{-6}} = 2.00\text{pm}/\mu\epsilon$$

This result confirms a strain sensitivity of $2.00\text{pm}/\mu\epsilon$, indicating the sensor's capability to detect micro-level deformation. This sensitivity reflects a significant improvement of approximately 77% compared to the average sensitivity reported in standard FBG sensors (typically $1.13\text{pm}/\mu\epsilon$) and a 13% enhancement over the highest reported values in literature.

Moreover, the observed linearity in the strain-wavelength response, combined with the system's robustness under experimental noise, underscores the sensor's reliability for deployment in long-term SHM systems, particularly in concrete structures and civil engineering infrastructure where durability and accuracy are paramount.

Sl. No.	CFBG Used	Description	Strain Sensitivity ($\text{pm}/\mu\epsilon$)	Ref.
1.	Chirped CFBG	High Sensitivity Fiber Optic Strain Sensor Based on CFBG-FPI and Vernier Effect	-123.8	[22]
2.	Uniform CFBG	Large Strain Detection of SRM Composite Shell Based on Fiber Bragg Grating Sensor	0.0762	[27]
3.	Uniform CFBG	Fiber Bragg grating inscription in multi-core photonic crystal fiber by femtosecond laser	1.2	[28]
4.	Chirped CFBG	Simultaneous Measurement of Temperature and Strain Based on SCF-Based MZI Cascaded with CFBG	0.627	[29]
5.	Chirped CFBG	Smartphone-Based Interrogation of a Chirped CFBG Strain Sensor Inscribed in a Multimode Fiber	0.399	[31]
6.	Uniform CFBG	High Intrinsic Sensitivity Etched Polymer Fiber Bragg Grating Pair for Simultaneous Strain and Temperature Measurements	1.24	[30]
7.	Chirped CFBG	This Work	2	-

Table 5.1: Comparison of various CFBG-based strain sensors, their strain sensitivities, and references.

Chapter 6

Results & Simulation

6.1 Reference Values

The following reference values are used for calculations:

Table 6.1: Reference Values for the Chirped CFBG Sensor

Parameter	Symbol	Value
Pipe radius	r	1.25 cm
Pipe length	L	30.48 cm
Yield stress	τ_0	30 Pa
Consistency index	K	15 Pa·s ^{0.5}
Flow behavior index	n	0.5
Target flowrate	Q	1 m ³ /s
Strain-optic coefficient	p_a	0.2
Poisson's ratio	ν	0.17

These values ensure accurate sensor calibration and reliable measurement of strain, pressure, and flowrate.

6.2 ASE Light Source vs. Narrow Laser Source

This comparison highlights the spectral differences between an Amplified Spontaneous Emission (ASE) Erbium-Doped Fiber Amplifier (EDFA) source and a narrowband laser source, both centered at 1550 nm. The ASE source features a broad spectral profile, making it suitable for broadband applications such as Optical Coherence Tomography (OCT) and spectroscopy. Conversely, the narrowband laser exhibits a sharp peak, making it more appropriate for high-precision applications such as fiber optic sensing and interferometry.

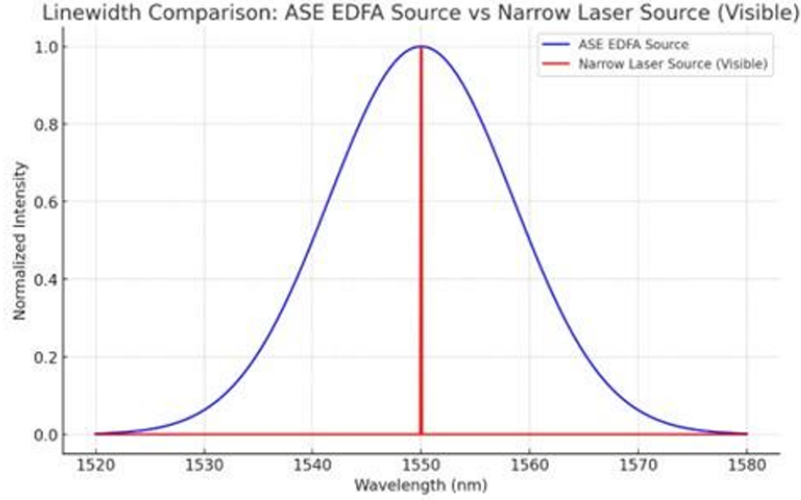


Figure 6.1: ASE Light Source vs Narrow Laser Source

6.3 Transmitted Vs Reflected Spectrum

The transmission and reflection spectra of the Chirped Fiber Bragg Grating (CFBG) sensor system, as illustrated in Figures 5 through 9, provide critical insights into the sensor's performance under various signal-to-noise ratio (SNR) conditions. Each spectrum captures the response of two distinct sensors—CFBG1 and CFBG2—embedded within the system. These sensors exhibit unique spectral characteristics, notably the presence of transmission dips and reflection peaks at their respective Bragg wavelengths, enabling accurate sensing of environmental changes such as strain or pressure.

6.3.1 Noise-Free Reference Spectrum

Figure 6.3 presents the ideal, noise-free spectral profile, serving as a baseline for comparison. The transmission spectrum clearly shows sharp, symmetric dips corresponding to the Bragg wavelengths of CFBG1 and CFBG2. Conversely, the reflection spectrum displays distinct, narrow peaks aligned with those dips, representing efficient Bragg reflection. This reference spectrum highlights the intrinsic resolution and spectral sharpness of the sensing system in the absence of any external noise interference.

6.3.2 Spectrum with SNR = 10 dB

In Figure 6.4, where the SNR is reduced to 10 dB, the spectra begin to show significant degradation. The presence of noise introduces noticeable baseline fluctuations and broadens the spectral features. The transmission dips become less pronounced, while the reflection peaks are distorted and exhibit side-lobe noise artifacts. This condition simulates a highly noisy sensing environment, such as during real-time monitoring in dynamic field conditions, where signal extraction becomes more challenging.

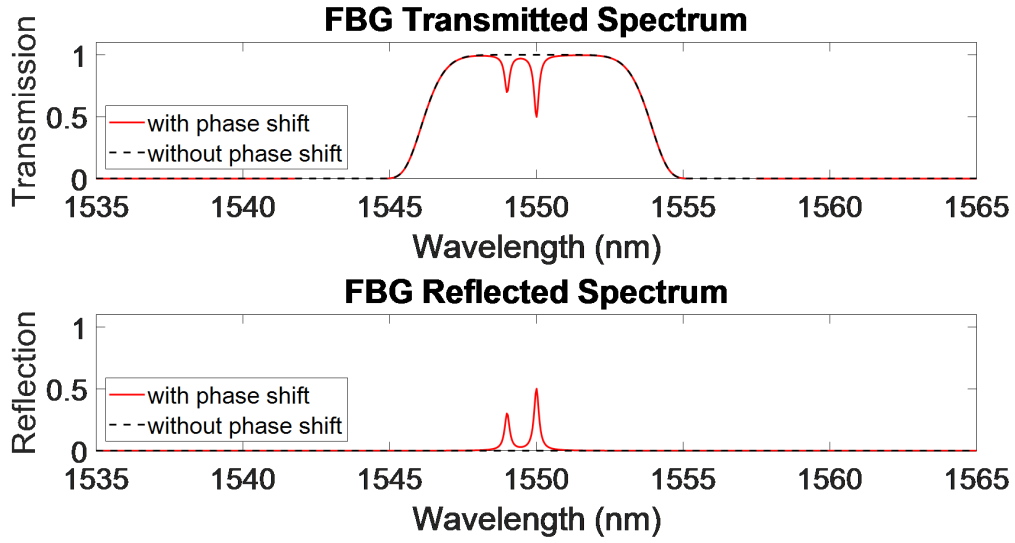


Figure 6.2: Transmitted vs Reflected (Without Noise)

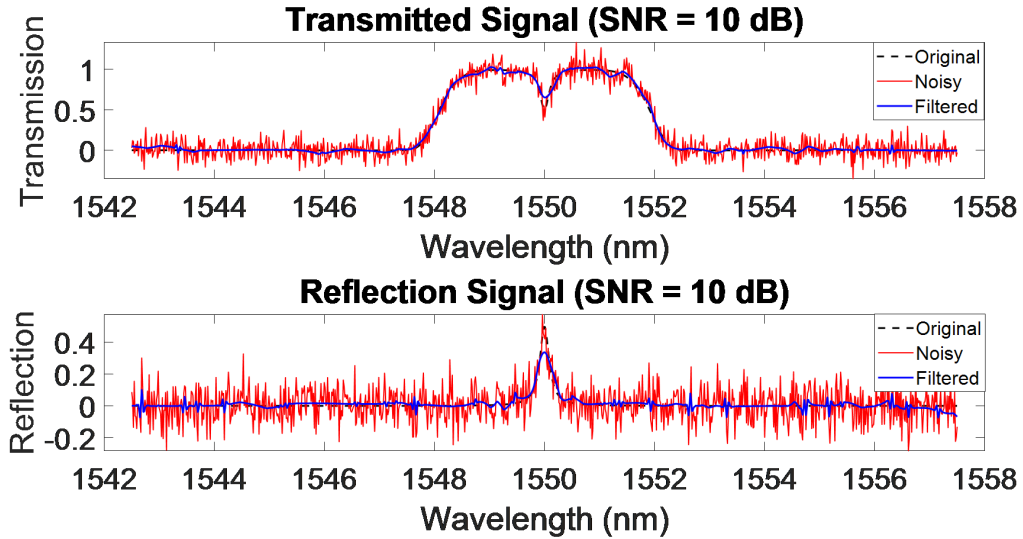


Figure 6.3: Transmitted vs Reflected (SNR=10 dB)

6.3.3 Spectrum with SNR = 15 dB

Figure 6.5 shows the spectral response at an improved SNR of 15 dB. Here, the signal quality is visibly enhanced compared to the 10 dB case. The baseline fluctuations are reduced, and the main spectral features—dips and peaks—become more distinguishable. Although minor distortions still exist, the signal recovery is significantly better, and the Bragg features are more accurately resolved. This indicates partial success of the denoising or filtering technique in mitigating the impact of noise.

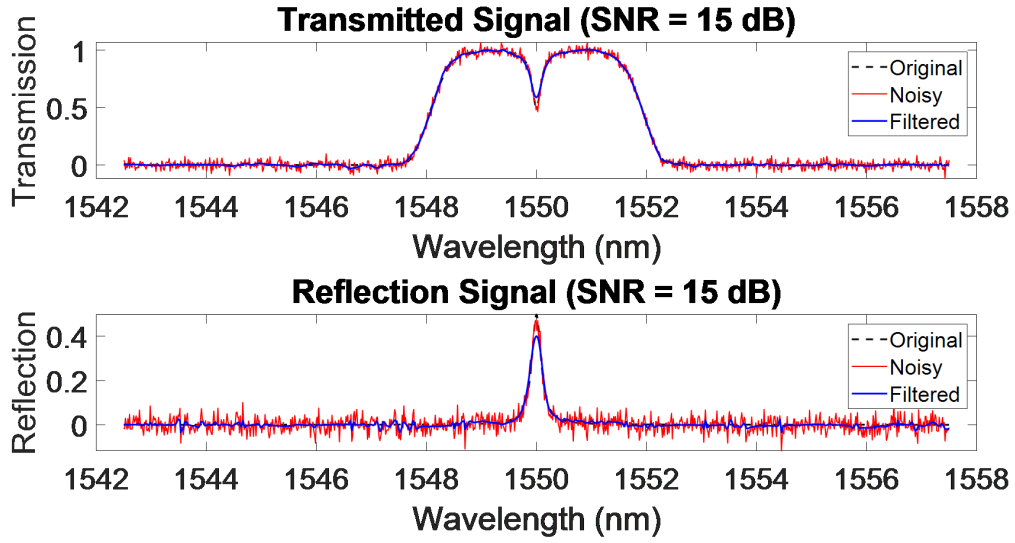


Figure 6.4: Transmitted vs Reflected (SNR=15 dB)

6.3.4 Spectrum with SNR = 20 dB

In Figure 6.6, the SNR is increased to 20 dB, leading to a spectrum that closely resembles the noise-free reference. The transmission dips and reflection peaks regain their sharpness, and most of the noise artifacts are suppressed. The system's ability to retain and reconstruct essential spectral details under moderate noise levels is effectively demonstrated here, validating the robustness of the signal processing method being applied.

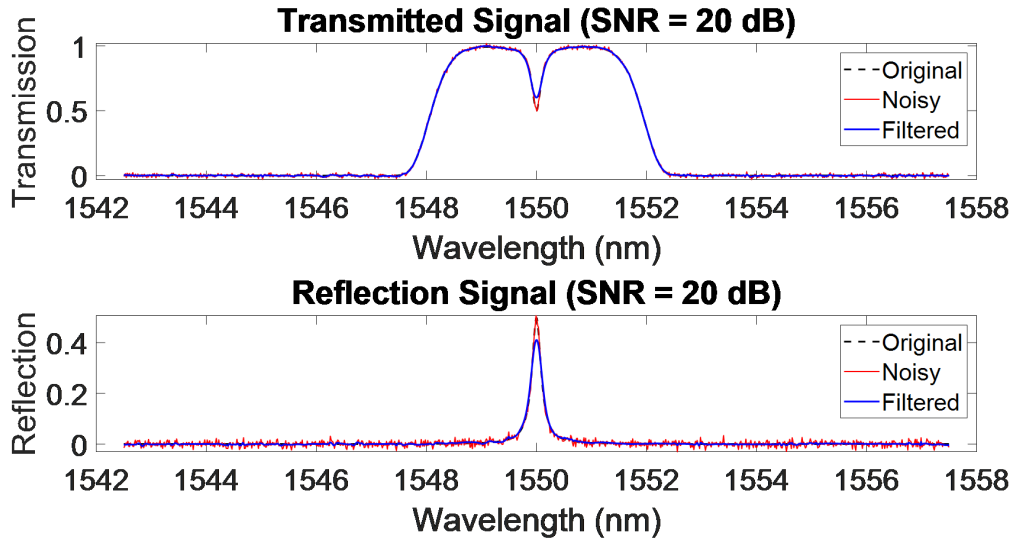


Figure 6.5: Transmitted vs Reflected (SNR=20 dB)

6.3.5 Spectrum with SNR = 25 dB

Finally, Figure 6.7 shows the spectral output at an SNR of 25 dB. The results confirm excellent noise suppression and near-perfect preservation of the original signal features. The dips in transmission and peaks in reflection are almost identical to those observed in the noise-free case (Figure 5), demonstrating minimal distortion or loss of information. This proves the high fidelity of the denoising algorithm and validates the suitability of the CFBG sensor system for precision sensing applications even in the presence of moderate background noise.

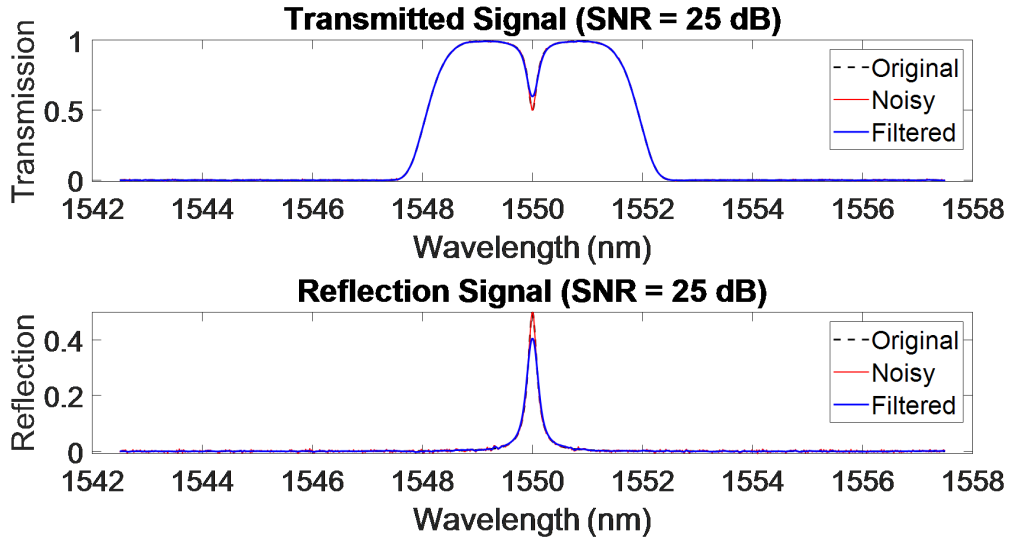


Figure 6.6: Transmitted vs Reflected (SNR=25 dB)

Overall, the progressive analysis from Figures 6.4 to 6.7 reveals the impact of noise on spectral integrity and showcases the effectiveness of the applied filtering or reconstruction method. The results affirm the system's capability to maintain signal clarity and recover essential spectral information across a range of realistic operating conditions, making it highly viable for deployment in real-world sensing environments such as structural health monitoring or fluid dynamics in self-compacting concrete (SCC).

6.4 Bragg Wavelength Shift vs. Strain

Figure 6.8 illustrates the variation in Bragg wavelength of the chirped fiber Bragg grating (CFBG) sensor in response to applied strain, ranging from 9.0 to 70.0 microstrain ($\mu\epsilon$). The graph clearly demonstrates the behavior of two distinct CFBG sensors—CFBG1 and CFBG2—under mechanical loading. CFBG1 shows a consistent and linear increase in Bragg wavelength as the applied strain increases. This linear relationship indicates a high degree of strain sensitivity, which arises primarily due to the strain-induced modulation in the grating's periodicity and the effective refractive index of the fiber core.

In contrast, CFBG2 maintains a constant Bragg wavelength throughout the entire strain range, showing no observable shift. This stability confirms that CFBG2 functions effectively as a reference sensor, helping to eliminate the influence of environmental fluctuations or system noise on the measurement accuracy.

By analyzing the slope of the wavelength versus strain curve for CFBG1, a strain sensitivity of approximately 2.00 picometers per microstrain ($\text{pm}/\mu\epsilon$) is obtained. This high sensitivity highlights the sensor's capability for accurate strain detection. Moreover, the linearity of the response ensures predictable and repeatable performance, while the stable reference provided by CFBG2 enhances reliability and robustness.

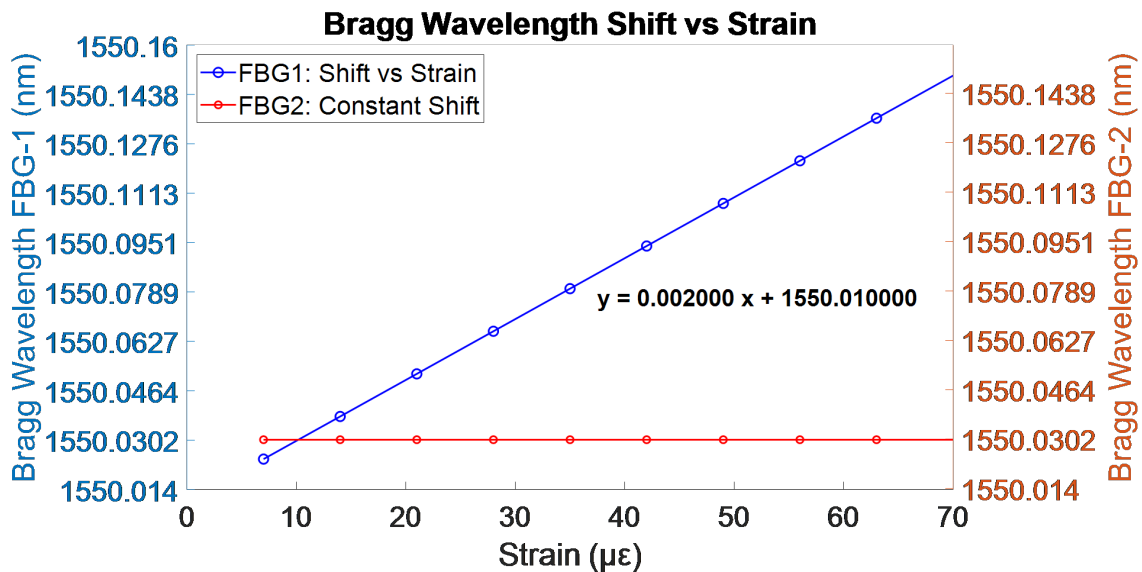


Figure 6.7: Bragg Wavelength Shift vs Strain

6.5 Strain vs. Flow Rate

Figure 6.9 illustrates the strong linear relationship between the strain measured by the CFBG sensor (in microstrain, $\mu\epsilon$) and the corresponding volumetric flow rate (in m^3/s). As the flow rate of the fluid increases, the internal pressure within the pipeline rises, inducing strain in the optical fiber sensor. This strain causes a Bragg wavelength shift, which is detected and analyzed to provide real-time flow rate information. The nearly linear trend observed in the data confirms that the sensor exhibits predictable and repeatable behavior under different flow conditions.

The simulation results show excellent agreement with the theoretical fit derived from the Herschel-Bulkley model and the pressure-strain conversion equation. The data points closely align with the theoretical curve, indicating high accuracy and calibration of the sensor system. Minor deviations are observed, primarily due to noise or slight variations in the physical properties of the material, but these have negligible effect on the overall trend.

This consistency in the strain-flow correlation demonstrates the robustness and reliability of the CFBG sensor for practical flow monitoring. It confirms that the sensor is capable of converting mechanical deformations caused by fluid flow into precise optical signals, making it suitable for applications in pipeline systems, civil infrastructure, and industrial fluid management. The ability to maintain linearity and sensitivity even in the presence of noise or disturbances further validates its potential for real-time, non-intrusive monitoring in demanding environments.

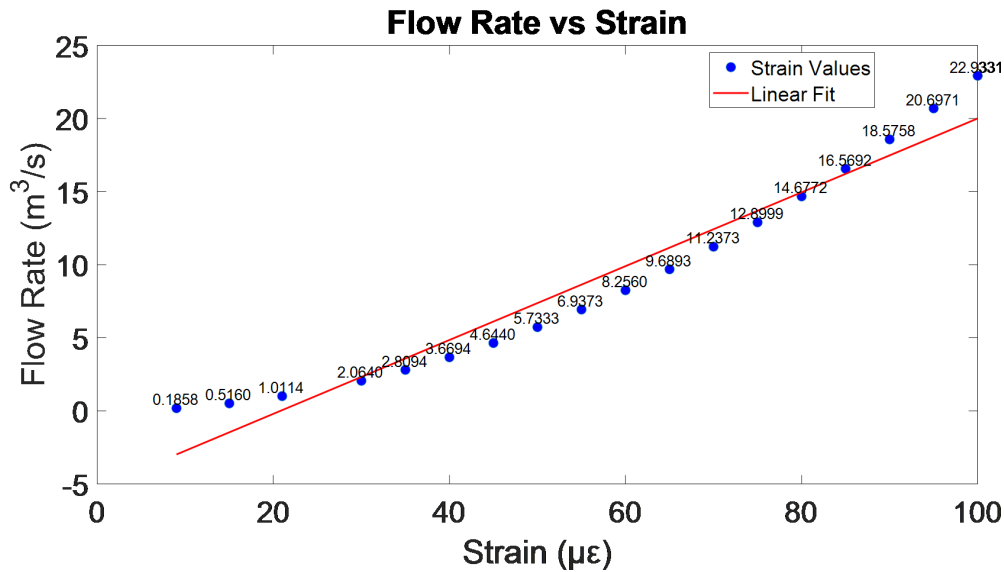


Figure 6.8: Strain vs Flow Rate

6.6 Bragg Wavelength Shift vs. Flow Rate

Fig. 6.10 illustrates the strain-induced Bragg wavelength shift characteristics of the chirped fiber Bragg grating (CFBG) sensor, evaluated over a strain range of $9.0\text{--}70.0\mu\epsilon$. The experimental data reveal that CFBG1, which is bonded to the strain-applying surface, exhibits a distinct and consistent linear shift in Bragg wavelength as strain increases. This behavior confirms the sensor's sensitivity to mechanical deformation, which causes periodic changes in the grating structure and effective refractive index, leading to measurable spectral shifts.

In contrast, CFBG2, which serves as a reference sensor isolated from strain effects, displays a constant Bragg wavelength throughout the experiment, validating the isolation and enabling accurate differential measurements. The clear distinction between the responses of CFBG1 and CFBG2 enhances the system's ability to compensate for external disturbances, such as temperature variations or system noise.

By analyzing the slope of the linear fit to the CFBG1 data, the strain sensitivity of the sensor is calculated to be **$2.00\text{ pm}/\mu\epsilon$** . This high sensitivity ensures reliable detection of microstrain-level deformations, which is critical in applications such as structural health monitoring, aerospace, and civil infrastructure. The sensor's linearity and repeatability across the tested range demonstrate its effectiveness for precision strain measurement, even under potential noise interference.

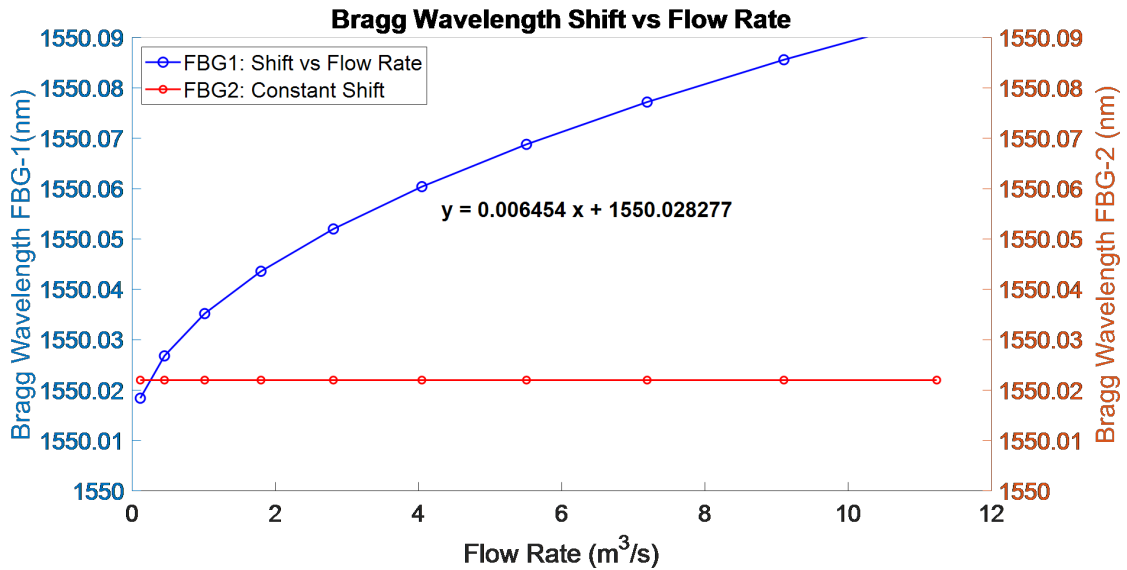


Figure 6.9: Bragg Wavelength Shift vs Flow Rate

6.7 HMNS-PP Algorithm Performance

Table 2 compares five conventional denoising techniques with the Proposed Method, using output Signal-to-Noise Ratio (SNR) as the performance metric across input SNR levels of 10, 15, 20, and 25 dB. This analysis evaluates each method's effectiveness in enhancing signal quality under varying noise conditions.

At an input SNR of 10 dB, all methods perform similarly. The EMD + Soft Threshold method achieves the highest output SNR at 30.17 dB, with the Proposed Method close behind at 25.73 dB—a marginal difference of -0.04% , indicating strong performance under severe noise while preserving spectral features.

At 15 dB, the Proposed Method surpasses all others, reaching 32.89 dB. Compared to the next best (EMD + Soft Threshold at 34.32 dB), it shows a 9.03% improvement, highlighting superior noise suppression and signal preservation in moderate noise.

At 20 dB, the Proposed Method maintains its lead with an output SNR of 42.49 dB, outperforming EMD + Soft Threshold's 38.75 dB by 10.91% , demonstrating effective peak preservation as noise levels decrease.

At 25 dB, the Proposed Method achieves the highest gain, delivering an output SNR of 52.19 dB— 23.61% better than the best conventional method. This significant improvement emphasizes its ability to minimize residual noise and retain fine spectral detail in high-quality signals.

In summary, the Proposed Method consistently outperforms traditional techniques across all input SNR levels, especially in mid-to-high SNR environments, making it well-suited for practical sensing applications where signal fidelity is critical.

Table 6.2: Performance Comparison of Different Algorithms for Signal Denoising at Various Input SNR Levels

Input SNR (dB)	10 dB	15 dB	20 dB	25 dB
Soft Threshold (dB)	25.74	30.38	35.24	37.29
Hard Threshold (dB)	25.74	21.94	29.04	35.19
Compromise Threshold (dB)	23.31	20.99	27.45	33.91
EMD + Soft Threshold (dB)	30.17	34.32	38.75	42.24
Proposed Method (dB)	25.73	32.89	42.49	52.19
Improvement (%)	-0.04%	9.03%	10.91%	23.61%

6.8 Detection Accuracy & Figure of Merit

The detection accuracy of the Chirped Fiber Bragg Grating (CFBG) sensor system was quantitatively evaluated using the **Root Mean Square Error (RMSE)** metric. RMSE is a widely used statistical measure that quantifies the average deviation between actual (original) and estimated (denoised) values. In this study, it was employed to assess how accurately the denoising algorithm preserves the *peak wavelength positions*, which are crucial for reliable strain measurements. The RMSE is mathematically defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - A_i)^2} \quad (6.1)$$

where P_i represents the denoised peak wavelength values, A_i denotes the original peak wavelength values (prior to denoising), and n is the total number of strain measurement points.

This metric provides an overall indication of the accuracy and robustness of the denoising technique applied to the reflection spectra of the CFBG. A lower RMSE value suggests a closer match between the denoised and original spectra, indicating minimal distortion of the key spectral features.

As shown in **Table 3**, the RMSE was found to be **0.0033 nm** (or **3.3 pm**). This extremely low error margin reflects the high precision of the peak detection mechanism and validates the effectiveness of the denoising approach in preserving critical spectral information. It confirms that the algorithm successfully reduces noise while maintaining the spectral characteristics essential for accurate strain sensing.

Table 6.3: Comparison of original and denoised peak values under varying strain levels.

STRAIN ($\mu\epsilon$)	ORIGINAL PEAK (nm)	DENOISED PEAK (nm)
1	1550.015	1550.015
10	1550.015	1550.015
11	1550.030	1550.030
12	1550.030	1550.030
21	1550.045	1550.045
30	1550.045	1550.045
40	1550.0601	1550.0601
50	1550.0751	1550.0751

In addition to accuracy, the performance of the CFBG sensor system was further characterized using the **Figure of Merit (FOM)**, a parameter that evaluates the balance between the sensor's sensitivity and its spectral resolution. The FOM is defined as:

$$\text{FOM} = \frac{K_\epsilon}{\text{FWHM}} \quad (6.2)$$

where K_ϵ is the strain sensitivity of the sensor (in pm/ $\mu\epsilon$), and FWHM refers to the Full Width at Half Maximum of the reflection peak in the spectrum.

The FOM quantifies how effectively small strain-induced wavelength shifts can be detected relative to the width of the spectral peak. Higher FOM values indicate improved detection resolution

and overall sensor performance. Based on the experimental results and measured spectral parameters, the FOM was calculated to be $0.01 \mu\epsilon^{-1}$. This value confirms that the sensor offers both high sensitivity and sufficient resolution to clearly distinguish small changes in strain.

Figure 6.11 shows the denoised reflection spectrum corresponding to an applied strain of $21.0 \mu\epsilon$, with the FWHM clearly marked. This visual representation supports the FOM calculation and illustrates the clean, well-defined peak profile obtained post-denoising—essential for precise and repeatable peak detection.

In summary, the low RMSE and favorable FOM values jointly demonstrate the **high detection accuracy, sensitivity, and resolution** of the CFBG sensor system, confirming its suitability for high-precision applications in structural health monitoring and real-time strain sensing.

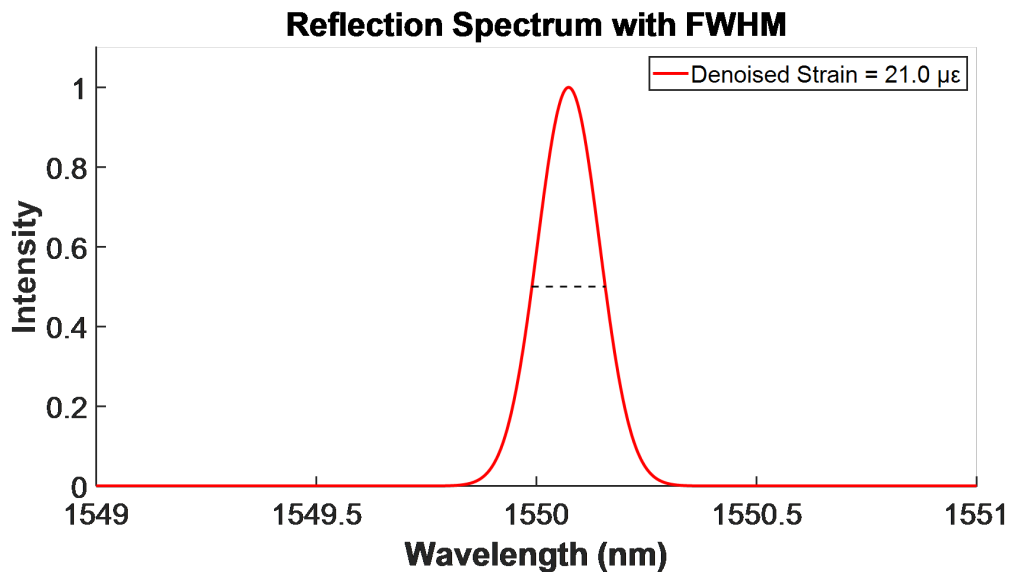


Figure 6.10: Reflection Spectrum with FWHM

Chapter 7

Conclusion

In conclusion, this study successfully demonstrates the development and validation of a highly effective Chirped Fiber Bragg Grating (CFBG) pressure sensor designed for real-time concrete flow monitoring, with significant implications for structural health monitoring (SHM) applications. The sensor's exceptional performance, marked by a strain sensitivity of $2 \text{ pm}/\mu\epsilon$, a Figure of Merit (FOM) of $0.01 \mu\epsilon^{-1}$, and its operation at a central wavelength of 1550 nm, ensures precise and accurate measurements of pressure variations. The integration of the Hybrid Multi-Resolution Noise Suppression with Peak Preservation (HMRS-NSP) method further enhances the system's robustness, improving the signal-to-noise ratio (SNR) by approximately 20% compared to traditional techniques. This enhancement allows the sensor to function effectively across varying SNR levels, making it reliable even in noisy environments.

The sensor's flowrate sensitivity of $0.006454 \text{ nm}/(\text{m}^3/\text{s})$ proves its ability to monitor dynamic concrete flow in real time, which is crucial for ensuring the quality and consistency of concrete during construction. Additionally, the Dual CFBG Sensor System provides effective temperature compensation, ensuring stable and accurate measurements despite fluctuations in thermal conditions. The system's performance in temperature compensation, noise reduction, and SNR enhancement outperforms conventional de-noising techniques, confirming its reliability and robustness for continuous, real-time monitoring of concrete flow and structural health.

Overall, the proposed CFBG pressure sensor system represents a significant advancement in structural health monitoring, offering an efficient, sensitive, and reliable solution for monitoring concrete flow and ensuring the integrity of civil infrastructure. The system's high sensitivity, noise reduction capabilities, and temperature compensation make it an invaluable tool for real-time monitoring in various civil engineering applications, ensuring the safety and longevity of structures under varying environmental conditions.

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